MILITARY SPECIFICATION
MULTIPLEXER TD-202( )/U

This specification is approved for use by Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for the manufacture and acceptance of the TD-202( )/U Multiplexer, a 12 or 24 channel radio combiner designed to accept PCM signals from associated tactical communication equipment, provide signal processing and input the PCM signals, suitably shaped and timed, to any appropriate radio set for transmission.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-S-92	Screw, Pan Head, Nylon
FF-N-836	Nut, Plain, Hex

MILITARY

MIL-P-11268	Parts, Materials, and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command, ATTN: DRSEL-PED-CM-DM, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal, DD Form 1426 (enclosed) appearing at the end of this document or by letter.

FSC-5805

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MILITARY (cont)

MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-S-19500	Semiconducting Device, General Specification For
MIL-P-55110	Printed Wiring Boards
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect
MIL-I-46058	Insulating Compound, Electrical (For coating printed circuit assemblies)
MIL-E-55585	Electronic Equipment and Parts, Packaging of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definitions of and Basic Requirements for Inclosures for Electric and Electronic Equipment
MIL-STD-220	Method of Insertion Loss Measurement
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirement for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interence Characteristics, Measurement of
MIL-STD-726	Packaging Requirement Codes
MIL-STD-781	Reliability Tests Exponential Distribution.
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-DL-527000	Multiplexer, TD-202( )/U
SC-DL-531012	Cable Assembly, RF, CG-1040
SC-DL-157329	Cable Assembly, RF, CG-1049
SC-DL-531015	Cable Assembly, Power, CX-1173( )/U
SC-DL-531018	Cable Assembly, Special Purpose, CG-2438( )/TCC
SC-GL-323241	Gages for TD-202( )/U
SM-B-423877	Breather Assembly
SM-B-423878	Cap, Breather

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DRAWINGS (cont)

SM-B-423879	Tube, Breather
SM-B-423880	Cover, Breather
SM-D-423893	Cover Assembly Case, Electronic Equipment
SM-D-423894	Cover Assembly
SM-C-423933	Latch Assembly
SM-D-423393	O-Ring
SC-DL-528640	Extender Panel
SM-B-527048	Power Supply 4A1/5A1
SM-B-527032	Panel Assy 4A6/5A6
SM-B-527033	Panel Assy 5A2
SM-B-529523	Panel Assy 4A8/5A8
SM-B-529524	Panel Assy 4A7/5A7
SM-B-529525	Panel Assy 5A3
SM-B-529526	Panel Assy 5A4
SM-B-529527	Panel Assy 5A5
SM-B-527087	Multiplexer TD-202 Test Requirement
SM-D-530670	Extender Cable Assembly

PUBLICATIONS

Department of the Army Technical Manual

TM 11-5805-367-12 Operator and Organizational Maintenance Manual Multiplexer TD-202( )/U (NSN 5805-00-884-2176). (Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification #6000A (Use UFC Code 34580)

(Application for copies should be addressed to the Western Railroad Association; Room 1150; 222 South Riverside Plaza; Chicago, Ill. 60606)

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services

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POST OFFICE DEPARTMENT

Parcel Post Regulations

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washinton, D.C. 20025)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification (Use NMFC Code 61700)

(Application for copies should be addressed to the National Classification Board; 1616 P Street, N.W.; Washington, D.C. 20036)

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and of Drawing and Data List SC-DL-527000 including the requirements for parts, materials and processes thereon.

3.3 Parts, materials and processes; general. (See 4.4) In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements herein, shall apply.

3.4 Signal Characteristics. Refer to Figure 10 for signal parameters.

3.4.1 PCM Signal Characteristics. (See 4.8.12). The pulsewidth of PCM signals at the PCM OUT 1 and PCM OUT 2 connectors when measured at the 50% amplitude points on the leading and lagging edges of the PCM signals shall be 1.736 microseconds for 12 channel (pulse rate 576 Khz), and 0.868 usec for 24 channel (pulse rate 1152 Khz). The rise and fall times shall be 20 to 120 nanoseconds (ns) when measured from the 10% to 90% amplitude points on the PCM signal. A digital "1" shall be a level between +0.2 and -0.2 volts and a digital "0" shall be a level between -1.8 and -2.2 volts. Ripple at either level shall be $\pm$ 0.3 volts. All output signals are specified for a load impedance of 91 ohms $\pm$ 10 %. The input impedance of those connectors that receive signals and the load impedance for those connectors that transmit signals shall be 91 ohms $\pm$ 10%.

3.4.2 Video Signal Characteristics. (See 4.8.13) The pulsewidth of the video signal output from the TO RADIO XMTR connector when measured at the 50% amplitude points on the leading and lagging edges of the PCM signals shall be 1.736 usec for 12 channels, and 0.868 usec for 24 channels. The rise and fall times shall be 160 to 280 ns when measured from the 10% to 90% amplitude points on the PCM signal. A digital "1" shall be level between 0.0 and -0.2 volts and a digital "0" shall be an adjustable level between -1.0 and -2.0 volts. All output signals are specified for a load impedance of 51 ohms $\pm$ 10%. Output levels shall be stable within $\pm$ 1 db. The input impedance of FROM RADIO XMTR shall be 51 ohms $\pm$ 10%.

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3.4.3 Timing Signal Characteristics. (See 4.8.14). Output signals at the TIMING OUT 1 and TIMING OUT 2 connectors shall be at a pulse rate of 576 Khz and when terminated in 91 ohms ±10% shall have a pulse peak 1.6 to 2.4 volts above a baseline of -1.8 to -2.2 volts. Ripple at the baseline shall be within ±0.5 volts. The rise and fall time shall be 20 to 120 ns when measured from the 10% to 90% amplitude points on the timing signal. The pulse width shall be 80 to 150 ns when measured at the 50% amplitude points on the leading and lagging edges of the timing signal. The 50% amplitude point on the leading edge of the TIMING OUT pulses shall lead the 50% point of the PCM OUT transitions by 0 to 80 ns for 12 to 24 channel separate operation, 0 to 80 ns for 24 channel repeater operation, and 868 to 948 ns for 24 channel interleaved operation. The input impedance of the TIMING IN connectors shall be 91 ohms ±10%.

3.4.4 Operating Characteristics.

3.4.4.1 Transmit Interleaving. (See 4.8.7). In the 24 channel mode, the equipment shall interleave two 12 channel PCM signals on a bit-by-bit basis such that the resulting pulse train shall be at a bit rate of 1152 Khz (twice the original bit rate). In case of failure of either 12 channel PCM input signal automatic switchover to emergency 12 channel operation shall occur. The two PCM signals will automatically be interleaved properly for any phase of PCM IN 2 signal.

3.4.4.2 Receive Framing and Separation. (4.8.9). The equipment shall separate a 24 channel interleaved PCM signal into its two 12 channel components. The PCM signal with the master (4Khz)address shall be presented at the PCM OUT 1 connector while the slave (2Khz) address PCM signal shall be presented at the PCM OUT 2 connector. The framing-in time for the 4Khz address searching circuits shall not be greater then 150 milliseconds (MSEC) for more than 10% of the tries while operating at or above the PCM threshold.

3.4.4.3. Binary Video Detection. (4.8.15, 4.8.16) In 12 channel operation the TD-202( )/U shall accept a PCM signal with timing from a transmitting multiplexer, sample and reshape the PCM signal, and provide a full width binary video signal to modulate a radio transmitter. The TD-202( )/U shall accept bandlimited gaussian shaped pulses as specified in TABLE II of a binary input video signal from a radio receiver, regenerate the PCM signal, and recover the timing. The Bit Error Rate (BER) shall be 10^{-6} or less with a maximum ratio of peak signal to RMS noise of 23 db for gaussian noise. The BER shall be 10^{-3} or less with a maximum ratio of peak signal to RMS noise of 18 db.

3.4.4.4 Biternary Video Detection. (4.8.17, 4.8.18) In 24 channel operation the TD-202( )/U shall accept two PCM signals from two transmitting multiplexers, interleave the two signals on a bit-by-bit basis, and provide a video signal to the radio transmitter consisting of full width binary pulses at twice the original bit rate. The TD-202( )/U shall also accept band-limited gaussian shaped pulses of a biternary input video signal from a radio receiver, regenerate the PCM signal, and recover the timing. The BER shall be 10^{-6} or less with a maximum ratio of peak signal to RMS noise of 28 db for gaussian noise. The BER shall be 10^{-3} or less with a maximum ratio of peak signal to RMS noise of 24 db.

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3.4.5 Order Wire.

3.4.5.1 Order Wire Characteristics. (See 4.8.10) The recovered order wire output level shall be adjustable from 0 to 1.5 volts RMS when loaded with a 600 ohm termination. The order wire recovery circuits shall be capable of providing -4dbm into 600 ohms with the FROM RADIO RCVR PCM signal set to 0.5v peak-to-peak and orderwire modulation 50 millivolts (mv) peak-to-peak. This range of adjustment shall be met when a 200 mv peak-to-peak 1 Khz sine wave is superimposed on a 0.75 volt peak-to-peak video pulse train at the FROM RADIO RCVR connector. The order wire clean up factor (order wire peak signal to RMS noise ratio after order wire recovery compared to order wire peak signal to RMS noise ratio before order wire recovery) shall be 20 db or better. The harmonic distortion caused by the order wire recovery circuits shall be not greater than 3% at a test frequency of 1 Khz and an output level of -4dbm. The radio order wire shall not be degraded by the order wire recovery circuit in the absence of an input video signal.

3.4.5.2 Service Condition Communication. (See 4.8.24 and 4.8.25) When tested in accordance with 4.8.24 and 4.8.25 voice communication shall be clear and intelligible. The BER shall be one error or less per minute.

3.4.6 Alarms.

3.4.6.1 Traffic Alarms. (See 4.8.11) The audible alarm shall sound and the TRAFFIC light shall illuminate after a one second delay following the loss of the PCM input signal. In addition the alarm condition shall disconnect the normally connected control lines to the order wire switchover circuit in the radio set. Depressing the BUZZER OFF switch shall silence the audible alarm. With the audible alarm silenced restoration of the PCM input signal shall cause the audible alarm to sound and the TRAFFIC light to extinguish. Depressing the BUZZER OFF switch shall again silence the audible alarm.

3.4.6.2 Change Air Filter Alarm. (See 4.12) The CHANGE AIR FILTER alarm shall operate when the temperature in the immediate area of the unregulated power supply reaches 71°C (160°F).

3.4.7 Power Supply.

3.4.7.1 AC Input. (See 4.8.19) The equipment shall not exhibit errors greater than 1 per minute, and shall successfully provide order wire communications and signalling when energized from 115 volts AC ±5%, frequency 47.5 and 63 Hz.

3.4.7.2 Output. (See 4.8.20) The 10, 4.5 and -4.5 DC power supply voltages shall be regulated to ±2.5%, with 10 millivolts RMS maximum ripple. The -12V DC power supply voltage shall be regulated to ±10%, with 30 millivolts max. RMS ripple. The foregoing characteristics shall apply over the AC input range cited in 3.4.7.1.

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3.4.7.3 Current and Voltage Range. (See 4.8.20) The voltage regulated current range for each power supply, and the minimum voltage range over which individual controls shall be able to adjust respective output voltages shall be as shown in Table I.

Table I

Nominal Regulated Output Voltage	Current Range (in milliampers)	Range of Manual Adjustment of Voltage
+10 Volts DC	225 to 275	9 to 11 Volts
+4.5 Volts DC	200 to 250	3.5 to 5.5 Volts
-4.5 Volts DC	700 to 900	-3.5 to -5.5 Volts
-12 Volts DC	80 to 120	Not Adjustable

3.4.7.4 Metering and Controls. (See 4.8.21) A hairline indication on the TEST ALIGN meter for the +10V, -4.5v and -12V SERV FAC switch positions shall occur for power supply voltages of +10 ±3%, +4.5 ±3%, and -12 ±3%. Likewise, a yellow tolerance band indication on the TEST ALIGN meter shall occur for power supply voltages falling within ±5% of their nominal vlaues. The controls for the manual adjustment of the +10V, +4.5V and -4.5V shall increase voltage output when rotated clockwise, and decrease voltage when rotated counterclockwise.

3.4.8 Test and Alignment.

3.4.8.1 TEST ALIGN Meter. (See 4.12) Placing the METER SELECT switch to the SERV FAC position shall connect the TEST ALIGN meter to the service selector switch on the side panel. Green band indications on the TEST ALIGN meter shall indicate presence of the signal being monitored. Absence of the signal being monitored shall cause an out of green band indication

3.4.8.2 TEST-OPR Switch. (See 4.8.23) With the TEST-OPR switch placed on TEST, the video output to the radio (TO RADIO XMTR) shall be routed through the band-limiting gaussian filter to the TEST OUT BNC connector for loop-back operation. The characteristics of the bandlimiting gaussian filter shall be as indicated in Table II.

Table II

Frequency (KHz)	Attenuation (db)
Below 116	Less then 1.0
166	1.5 ± 0.25
235	3.0 ± 0.25
332	6.0 ± 0.50
470	12.0 ± 0.50
667	24 ± 1.0
944	48 ± 2.0

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3.4.8.3 TO RADIO XMTR Alignment. (See 4.8.22) The level of the output signal at the TO RADIO XMTR connector shall be adjustable from 1.0 volt to 2.0 volts peak-to-peak utilizing the OL (output level) control located on the 5A2 panel. With the METER SELECT switch in the TO RADIO XMTR position the TEST ALIGN meter shall provide a hairline indication for a 1.5 volt ±0.1 volt peak-to-peak output signal at the TO RADIO XMTR connector.

3.4.8.4 FROM RADIO RCVR Alignment. (See 4.12) With the METER SELECT switch in the FROM RADIO XMTR position input signals within the range of 0.5 volt to 3.0 volts at the FROM RADIO RCVR connector shall be adjustable to a hairline indication on the TEST ALIGN meter. Adjustment shall be accomplished utilizing the RL (receive level) control located on the 5A3 panel. The most clockwise position of the RL control shall correspond to a hairline reading with the smallest input signal.

3.4.8.5 Center Level (CL) Alignment. (See 4.12) With the METER SELECT switch in the SERV FAC position and the SERV SEL switch in the C position the correct CL, corresponding to a maximum (peak) indication on the TEST ALIGN meter, shall be obtainble by adjusting the CL control located on the 5A3 panel.

3.4.8.6 ORDER WIRE LEVEL Alignment. (See 4.12) With the METER SEL switch in the SERV FAC position and the SERV SEL switch in the O position the ORDER WIRE LEVEL control shall be capable of adjusting the order wire output level. A hairline indication on the TEST ALIGN meter shall correspond to a -4dbm output at the ORDER WIRE connector (600 ohm termination). A hairline indication shall be observed for all but extreme (90 percent zeros or ones) pulse modulation patterns. Clockwise rotation of the Order Wire Level control shall increase the output level of the recovered order wire.

3.4.8.7 Positive and Negative Level Adjustment. (See 4.12) The positive level (PL) and negative level (NL) controls located on the 5A3 panel shall be capable of adjusting the biternary decision levels in the PCM detector to the proper respective peak video levels (after clamping).

3.4.8.8 Binary and Biternary Decision Timing Phase Adjustment. (See 4.12) The phase adjustment (PA) for the binary detection phase located on the 5A4 panel shall have sufficient range to align the decision timing to the center of the binary pulses. The PA adjustment for the biternary detection phase on the 5A5 panel shall have sufficient range to align the decision timing to the "zero" cross-over time of the pulse transitions.

3.5 Preconditioning.

3.5.1 Electrical Preconditioning (Burn-in). (See 4.7.1) Prior to submission to the specified Quality Conformance Inspections, and subsequent to the bounce preconditioning of 3.5.2 each assembled Multiplexer TD-202( )/U consisting of chassis assembly, all panel assemblies, and the power supply shall be electrically preconditioned for a minimum of 48 hours. The last

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24 hours of electrical preconditioning shall be failure free. The maximum preconditioning time allowable for any unit, assembly or subassembly to be delivered on contract shall be 168 hours. Each fabricated TD-202( )/U shall be subjected to this process. All end equipments failing Group A inspection (See 4.5.1) may be repaired and tested to the end equipment Group A requirement then accepted and delivered without repeat of electrical pre-conditioning. Repair actions involving panel assemblies shall be accomplished through use of preconditioned panels.

3.5.2 Bounce Preconditioning. (See 4.7.2) Each TD-202( )/U assembly shall meet the requirements herein without subsequent processing after submission to the Bounce Preconditioning of 4.7.2. Each fabricated TD-202( )/U shall be subjected to this process.

3.6 Service Conditions. (See 4.6) The TD-202( )/U shall meet the Service conditions listed below. Where a test is referenced, meeting the test shall be considered as compliance with the requirements.

3.6.1 Temperature. The following low and high temperature requirements shall be met:

3.6.1.1 Low. (See 4.6.1.1) The equipment shall be operated during and after testing as specified in 4.6.1.1, and shall meet the measurements of 4.8.24 at room ambient temperature and at -25°F. The equipment shall withstand exposure to a non-operating temperature of -65°F.

3.6.1.2 High. (See 4.6.1.2) During and after the testing specified in 4.6.1.2, the equipment shall operate and meet the requirements of 4.8.24 at room ambient and +125°F with simultaneous solar radiation of 360 BTU/sq. ft/hr. The equipment shall withstand exposure to a non-operating temperature of +154°F.

3.6.2 Moisture resistance (humidity). (See 4.6.2) During and after testing as specified in 4.6.2 the equipment shall meet the requirements for the measurements as specified in 4.8.24. The equipment shall exhibit no physical damage such as corrosion, rust, blistering, swelling or deterioration of parts and materials.

3.6.3 Low pressure (altitude) (See 4.6.3) During and after testing as specified in 4.6.3 the equipment shall be operable and meet the requirements of 4.8.25 at altitudes of 15,000 ft above sea level and shall withstand transportation of altitudes at 50,000 feet above sea level.

3.6.4 Vibration, Bounce and Shock.

3.6.4.1 Vibration, internal (See 4.6.4). During and after testing as specified in 4.6.4, the equipment shall meet full specification performance for the measurements specified in 4.8.25.

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3.6.4.2 Bounce (vehicular) (see 4.6.5). The equipment shall meet the requirements for the tests specified in 4.8.25 after subjection to the test specified in 4.6.5.

3.6.4.3 Shock (bench handling). (See 4.6.6) The equipment in its servicing condition shall meet the requirements for the tests specified in 4.8.25 after subjection to the test specified in 4.6.6. There shall be no evidence of physical damage except minor surface abrasion.

3.6.5 Fungus. (See 4.6.10). After the fungus test of 4.6.10, the equipment shall be visually examined, using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, or sparse microbial growth (see 6.3) as evidence by growth colonization, which includes branching and sporulation on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.3) shall not be included in this evaluation.

3.6.6 Dripproof. (see 4.6.7). The equipment in normal operating condition shall meet the requirements of 4.8.25 when tested in accordance with 4.6.7 with the cover assembly installed (See 4.14). The equipment shall not leak or show evidence of leakage at the conclusion of the test.

3.6.7 Salt Fog (see 4.6.8). After the salt fog test of 4.6.8, the equipment, when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surface; or exfoliation (see 6.3) of metal. Electrical operation is not required.

3.6.8 Dust (Fine Sand) (see 4.6.9). After testing as specified in 4.6.9, the equipment shall meet the requirements for the tests specified in 4.8.25. Except for exterior paint damage on the case or front panel, the equipment shall exhibit no electrical or mechanical damage.

3.7 Finish, Protective. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings (see 4.4).

3.8 Marking. Marking shall conform to Specification MIL-M-13231. (See 4.4). Front panel marking shall be Group I as described in that specification.

3.8.1 Visibility. Wherever practicable parts shall be mounted so that their identification markings will be readily visible with minimum disassembly of the equipment.

3.9 Serial Numbers. (see 4.4 Marking). Each Multiplexer TD-202 ( )/U shall be serial numbered.

3.10 Electromagnetic Interference Characteristics. (see 4.9). The equipment shall meet the electromagnetic interference emission and susceptibility

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requirements of MIL-STD-461A, Notice 4, Table A-1, for Class CE equipment except that only tests CE04, CS02, CS06, RE02, RE02.1, and RS03 are applicable. Tests and test methods shall be as specified in MIL-STD-462, Notice 3.

3.11 Interchangeability (see 4.10). Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.12 Panel Assemblies (Electrical Test). (see 4.12). The TD-202( )/U and panel assemblies listed below shall meet the requirements of their respective test requirement drawings as specified in 4.12.

Multiplexer TD-202( )/U (TRS Test and Alignment Requirements) SM-B-527087

Power Supply 4A1/5A1	SM-B-527048	Panel Assy 4A7/5A7	SM-B-529524
Panel Assy 4A6/5A6	SM-B-527032	Panel Assy 5A3	SM-B-529525
Panel Assy 5A2	SM-B-527033	Panel Assy 5A4	SM-B-529526
Panel Assy 4A8/5A8	SM-B-529523	Panel Assy 5A5	SM-B-529527

3.13 Safety engineering.

3.13.1 Personnel hazards (see 4.11). Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

3.13.2 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spark gaps, lenses).

3.14 Operational test (see 4.8.6) Each fabricated and tested Multiplexer TD-202( )/U shall be subjected to a Final Operational Test in accordance with 4.8.6 after all other inspections and tests are completed to determine that the TD-202( )/U is operational prior to packaging.

3.15 Workmanship (see 4.4 and 4.11). The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack.
- Protection.
- Clearance.
- Splicing
- Connections, general.
- Grounding, general.
- Shielding on wire and cable.

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Soldering and brazing.
Cleaning.
General requirements for securing of parts.
Cleaning prior to welding.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.

3.16 Printed wiring assemblies (see 4.4). Printed wiring assemblies shall be in accordance with the applicable paragraph in MIL-P-11268 and shall be conformally coated in accordance with MIL-STD-275 with a coating material that conforms to MIL-I-46058.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Test equipment and facilities for Government Inspection shall be made available by the contractor.

4.2 Classification of Inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3) - does not include preparation for delivery.

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.13).

4.3 First Article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First Article units. The contractor shall furnish six (6) first article units of the complete Multiplexer TD-202( )/U.

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4.3.2 First Article inspection. The first article inspection shall consist of inspections specified in Table III, and shall be performed in the order specified herein.

4.3.3 First Article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement in the contract or purchase order.

Item	Where Required
Parts, Materials, and processes; General	3.3
Finish	3.7
Marking	3.8
Microelectronic Integrated Circuits	3.9
Workmanship	3.15
Printed Wiring Assemblies	3.16

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspections specified in 4.4, 4.7, and 4.5.1 through 4.5.4 in that order. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specified requirements.

4.5.1 Group A inspection. Each unit identified in Table III (and each sub-assembly and assembly on contract or purchase order) shall be inspected for conformance to the inspections specified in Table IV. The order of inspection of the subassemblies and assemblies and units (end equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in Table IV shall be subject to a 100% visual/ mechanical and electrical inspection by the contractor for conformance with the inspection requirements also provide in Table IV.

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b. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using general inspection levels and Acceptable Quality Level's (AQL) indicated in Table IV (Quality Assurance Verification Inspection). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies from the discrete lots which have passed the above Quality Assurance Verification Inspection shall be assembled into the end equipments which shall then be subjected by the contractor to the preconditioning requirements of 3.5 and a 100% visual/mechanical and electrical inspection in accordance with Table IV.

NOTE: Assemblies are not required to undergo furhter visual/mechanical and electrical inspection as part of the contractor's 100% Group A Inspection of the end equipment.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQL's indicated in Table IV.

NOTE: Assemblies and subassemblies are not required to undergo further visual mechanical and electrical inspection as part of the above contractor's sampling inspection of the end equipment.

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TABLE III - FIRST ARTICLE INSPECTION

Inspections	Rqt. Para.	Test Para.	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6
1. Inspections covered by subsidiary documents		4.4	Inspection to be performed on all units					
2. Bounce preconditioning	3.5.2	4.7.2	Inspection to be performed on all units					
3. Electrical preconditioning	3.5.1	4.7.1	Inspection to be performed on all units					
4. Group A Inspection		Table IV	Inspection to be performed on all units					
5. Group B Inspection		Table V	Inspection to be performed on all units					
6. Group C Inspection		Table VI						
Low Temperature	3.6.1.1	4.6.1.1					1	1
High Temperature	3.6.1.2	4.6.1.2					2	2
Humidity	3.6.2	4.6.2					3	3
Low Pressure (altitude)	3.6.3	4.6.3		1				
Vibration (internal)	3.6.4.1	4.6.4	1					
Bounce (vehicular)	3.6.4.2	4.6.5	2					
Shock (bench handling)	3.6.4.3	4.6.6				3		
Fungus (see Note c)	3.6.5	4.6.10			1			
Dripproof	3.6.6	4.6.7				2		
Salt Fog (see Note c)	3.6.7	4.6.8		3				
Dust	3.6.8	4.6.9		2				
EMI	3.10	4.9				1		

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NOTES:

a. The inspections 1 through 5, in the order shown, shall be performed on all First Article units before subjecting these units to any other inspection requirements in the table.

b. The numbers in the unit columns in the table covered by Inspection. numbered 6 specify the order of inspections for the indicated unit.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

TABLE IV GROUP A INSPECTION

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual-mechanical				
TD-202( )/U Case & Covers	3.15	4.11	2.5*	10.0*
Panel Assy 4A6/5A6	3.15	4.11	1.0*	4.0*
Panel Assy 5A2	3.15	4.11	1.0*	4.0*
Panel Assy 4A8/5A8	3.15	4.11	1.0*	4.0*
Panel Assy 4A7/5A7	3.15	4.11	1.0*	4.0*
Panel Assy 5A3	3.15	4.11	1.0*	4.0*
Panel Assy 5A4	3.15	4.11	1.0*	4.0*
Power Supply 4A/1/5A1	3.15	4.11	1.0*	4.0*
Panel Assy 5A5	3.15	4.11	1.0*	4.0*

*Defects per hundred units.

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TABLE IV GROUP A INSPECTION (cont'd)

Inspection	Req Para	Insp Para	Ref Fig	AQL Major	AQL Minor
Electrical Performance (Including spare panel assy)					
Power Supply 4A1/5A1	3.12	4.12		1.0%	**
Panel Assy 5A5	3.12	4.12		1.0%	**
Panel Assy 5A3	3.12	4.12		1.0%	**
Panel Assy 4A7/5A7	3.12	4.12		1.0%	**
Panel Assy 4A8/5A8	3.12	4.12		1.0%	**
Panel Assy 5A2	3.12	4.12		1.0%	**
Panel Assy 4A6/5A6	3.12	4.12		1.0%	**
Multiplexer TD-202( )/U	3.12	4.12		1.0%	**
Electrical					
Operational Test	3.14	4.8.6	1	1.0%	**
24 Channel Switchover Test	3.4.4.1	4.8.7	2	1.0%	**
Automatic Phasing Test	3.4.4.1	4.8.8	3	1.0%	**
Separation Test	3.4.4.2	4.8.9	4	1.0%	**
To RADIO XMTR Alignment	3.4.8.4	4.8.22	15	1.0%	**

** All electrical defects are major

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government with the Operational Test performed last.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table V and to the special inspection levels of Table I of MIL-STD-105.

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TABLE V GROUP B INSPECTION

Inspection	Req Para	Ref Fig	Insp Para	AQL
Orderwire Recovery Test	3.4.5.1	5	4.8.10	1%
Traffic Alarm Test	3.4.6.1	6	4.8.11	1%
PCM Signal Test	3.4.1	7	4.8.12	4%
Video Signal Test	3.4.2	8	4.8.13	4%
Timing Signal Test	3.4.3	9	4.8.14	4%
Binary Video Detection Test	3.4.4.3	10	4.8.15	4%
		10	4.8.16	4%
Biternary Video Detection Test	3.4.4.4	11	4.8.17	4%
		11	4.8.18	4%
AC Input Test	3.4.7.1	12	4.8.19	4%
Power Supply Output Test	3.4.7.2	12	4.8.19	4%
	3.4.7.3	13	4.8.20	4%
Metering Test	3.4.7.4	14	4.8.21	4%
Gaussian LP Filter Test	3.4.8.2	16	4.8.23	4%
Interchangeability	3.11		4.10	4%

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4.5.2.1 Group B sampling plans. The Group B AQL shall be as shown in Table V and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table VI. Samples shall be selected in accordance with 4.5.3.1.

TABLE VI GROUP C INSPECTION

Inspection	Req Para	Ref Fig	Insp Para
Sub-Group 1			
Low Temperature	3.6.1.1	17	4.6.1.1, 4.8.24
High Temperature	3.6.1.2	17	4.6.1.2, 4.8.24
Moisture Resistance (Humidity)	3.6.2	17	4.6.2, 4.8.24
Sub-Group 2			
Dripproof	3.6.6	17	4.6.7, 4.8.25
Vibration, internal	3.6.4.1	17	4.6.4, 4.8.25
Bounce, vehicular	3.6.4.2	17	4.6.5, 4.8.25
Shock (Bench Handling)	3.6.4.3	17	4.6.6, 4.8.25
EMI	3.10	18	4.9
Sub-Group 3			
Low Pressure (altitude)	3.6.3	19	4.6.3, 4.8.25
Dust (Fine Sand)	3.6.8	19	4.6.9 (See note 1)
Salt Fog	3.6.7		4.6.8 (See Note 1)
Fungus	3.6.5		4.6.10(See Note 1)

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried and refurbished if necessary, before proceeding with subsequent tests.

4.5.3.1 Sampling for Group C inspection

4.5.3.1.1 Sug-group 1. For this subgroup, two (2) units shall be selected from the first production lot. For subsequent Group C inspections, one unit shall be selected every month. Each unit shall be subjected to Temperature and Humidity testing in that order.

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4.5.3.1.2 Sub-group 2. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspections, two units shall be selected every third month. One unit shall be subjected to EMI, Dripproof, and Shock, (Bench Handling) testing; and one unit shall be subjected to Vibration (internal), and Bounce Vehicular testing with each unit subjected to testing in the order given.

4.5.3.1.3 Sub-group 3. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspections, two units shall be selected every fourth month. One unit shall be subjected to Altitude (Elevation), Dust and Salt Fog testing, in that order. One unit shall be subjected to Fungus testing.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.6 Service condition tests. (see 3.6) The following service condition tests shall be performed. Since more than one unit is available, some tests may be run concurrently with Government approval by the contracting officer or his designated representative. After completion of the Fungus and Salt Fog tests, the equipment must be thoroughly washed, cleaned, dried and refurbished, if necessary before proceeding with subsequent tests.

4.6.1 Temperature Tests.

4.6.1.1 Low temperature. (see 3.6.1.1) The equipment shall be subjected to Test Method 502.1, Procedure I of MIL-STD-810. Storage temperature shall be -65°F and operation temperature shall be -25°F. Storage time shall be 24 hours. At operation temperature, tests shall start within 5 minutes after ac power application. A pretest performance record is required and is specified in 4.8.24. Performance checks per 4.8.24 shall be made as specified in Test Method 502.1 during tests.

4.6.1.2 High temperature. (see 3.6.1.2) The equipment shall be subjected to Test Method 501.1, Procedure II of MIL-STD-810 with the following exceptions:

Step 4 Utilize 68°C(155°F) as the internal chamber temperature rather than 71°C (160°F).

Step 8 Incorporate the following in the existing step: "Solar radiation of the intensity of 360 BTU/sq. ft./hr. shall be maintained for

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4 hours and then the equipment shall be tested in accordance with paragraph 4.8.24. The highest operating temperature is 52°C + 3°-0°C (125°F + 5°-0°F) not including the effects of solar radiation. Performance tests per 4.8.24 shall be made during tests as specified in this procedure."

4.6.2 Moisture Resistance test (humidity) (see 3.6.2.) The equipment with the power supply, top, rear, and panel covers removed shall be subjected to Test Method 507.1, Procedure II of MIL-STD-810. Test measurements per 4.8.24 shall be taken during the 24 to 28 hr period and the last 5 hr period during each 48 hour cycle.

4.6.2.1 Failure. If any equipment fails to meet the requirements for the tests specified in 4.8.24 during cycling, it shall be realigned or readjusted once. If the equipment then fails or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.8.24 it does not pass the test.

4.6.3 Low pressure (Altitude). (see 3.6.3.) The equipment shall be subjected to the following test:

a. The equipment shall be placed in the test chamber. Prior to testing, the equipment shall meet the requirements for the tests specified in 4.8.25 except that auxiliary test equipment need not be inclosed in the chamber. A temperature of 73° ±18°F shall be maintained during the entire test.

b. Reduce the chamber internal pressure to 87.376 mm of Hg (3.44 in of Hg or 50,000 feet above sea level). The rate of pressure change may be the maximum attainable by the chamber. Maintain the chamber pressure for a period of not less than one hour.

c. Increase the chamber pressure to 429.26 mm of Hg (16.9 in. of Hg or 15,000 feet above sea level) and then perform the test of 4.8.25 and record results.

d. With the equipment not operating, return the chamber to standard ambient conditions and perform the test of 4.8.25 and visually inspect the equipment.

e. During and after the test, the equipment shall meet the requirements for the test of 4.8.25 and shall not exhibit any electrical or mechanical degradation which could in any manner prevent the equipment from meeting any requirements of this specification.

4.6.4 Vibration test (internal) (see 3.6.4.1).

a. Resonant frequencies of the equipment shall be determined by varying the frequency of applied vibration slowly through the specified range at reduced tests levels but with sufficient amplitude to excite the item.

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b. The test item shall be attached to the vibration exciter table by its normal mounting means or by means of a rigid fixture capable of transmitting the vibration conditions. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of undesirable responses in the test. The TD-202/U shall be mounted on the vibration table positioned in its normal operating plane. The top and rear covers shall be removed. The card cell cover and connector covers shall be in place. The vibration shall be in a plane perpendicular to the plane of the printed circuit boards.

c. The test item shall be operated during the vibration resonance search so that functional effects caused by this test may be evaluated.

d. The test level shall be applied over a frequency range of 10 to 55 Hz and at a vibratory displacement of not less than 0.030 inches, double amplitude. The vibratory frequency shall be changed in discrete steps of 1 Hz and maintained at least 10 seconds at each frequency. Where possible, fixturing shall be such that reasonances within the test item can be observed and measured. The equipment shall have no reasonances in the frequency range of 10 to 55 Hz that exceed twice the amplitude of applied vibration.

e. The equipment shall meet pretest and post test performance requirements in accordance with paragraph 4.8.25.

4.6.5 Bounce (vehicular) test (see 3.6.4.2). The equipment in its normal operating configuration with connector covers installed shall be subjected to Test Method 5.4.2, Procedure IX, Part 2 of MIL-STD-810. At the conclusion of test, the equipment shall meet full specified performance when tested in accordance with 4.8.25. Pretest and post test data per 4.8.25 is required. The Card Cell Cover shall be installed during all drops. All other covers shall be removed.

4.6.6 Shock (bench handling) test (see 3.6.4.3) The equipment with connector covers installed shall be subjected to Test Method 516.2, Procedure V of MIL-STD-810. At the conclusion of test, the equipment shall meet full specified performance when tested in accordance with 4.8.25. Pretest and post test data per 4.8.25 is required. The Card Cell Cover shall be installed during all drops. All other covers shall be removed.

4.6.7 Dripproof test (see 3.6.6). The equipment shall be tested per MIL-STD-108, 4.3(b) Drip Test, with the Cover Assembly installed (see 4.14), except as follows:

a. The equipment shall be in its normal operating configuration.

b. In lieu of equipment orientation 15° from the vertical, drops of liquid shall be falling at an angle of 15° from the vertical when the equipment is mounted in its normal position.

c. Immediately following the drip test, the equipment shall meet the full specified performance when tested in accordance with 4.8.25.

4.6.8 Salt Fog Test (see 3.6.7) The equipment in its normal operating configuration with connector covers and Cover Assembly (See 4.14) installed shall be subjected to the Salt Fog test specified in Method 509.1, Procedure I, of MIL-STD-810. No corrosion products shall be deposited on the test item by the facility.

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At the conclusion of the test, the equipment shall be examined for evidence of visible corrosion.

4.6.9 Dust Fine Sand test (see 3.6.8). The equipment, in its normal operating configuration, with connector covers and Cover Assembly (see 4.14) installed shall be subjected to Test Methoe 510.1, Procedure I of MIL-STD-810. The equipment shall not be operated during the test. Pretest and post test data is required in accordance with 4.8.25.

4.6.10 Fungus test (see 3.6.5). The equipment in its normal operating configuration with connector covers installed shall be subjected to the fungus test specified in Method 508.1, Procedure 1 of MIL-STD-810. There shall be abundant growth colonization (6.3) on 50% or more of the are of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.6.5 shall constitute failure of this test.

4.7 Preconditioning. (see 3.5)

4.7.1 Electrical preconditioning inspection (burn-in) (see 3.5.1). Electrical preconditioning inspection shall meet the requirements of 3.5.1.

4.7.1.1 Test Details.

Electrical preconditioning (burn-in). Test level B of MIL-STD-781 shall apply for temperature and vibration requirements.

a. During testing, the equipment shall be mounted on a vibration table (in its normal operating configuration), and positioned in its normal operating plane with all covers in place. Vibration shall be in a plane perpendicular to the plane of the printed circuit boards. During all operating periods, the equipment under test shall be electrically connected per Figure 1. The equipment shall be turned on and operated in the 24 channel mode for 23 hours and then turned off for 1 hour. The next 24 hour period, 23 hours operating and 1 hour off, shall be in the 12 channel mode as indicated in Fig 1. Time expended to perform the Operational tests shall be considered part of the "ON" cycle of the "ON-OFF" cycle.

b. The chassis, plug-in printed circuit panel assemblies and power supplies shall be considered to have been successfully electrically preconditioned when they have individually passed a cumulative 48 hour operating preconditioning period without a failure during the last 24 hour period. If during the last 24 hour operating preconditioning period a plug-in panel assembly, power supply, or chassis fails, it may be replaced with a preconditioned panel, power supply, or chassis and the preconditioning continued without the restart of a new 48 hour preconditioning period. A plug-in panel assembly, power supply or chassis failing operating preconditiong shall, after repair, be resubmitted to and pass the entire 48 hour operation preconditioning period prior to submission to subsequent quality conformance inspection.

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c. The unit chassis shall have no less than 48 hours of operational preconditioning nor more than 168 hours of operational preconditioning (exclusive of production test time) when used as a vehicle for preconditioning power supplies and plug-in assemblies (including all spares).

4.7.1.2 Marking of electrically preconditioned units. All plug-in panel assemblies, power supplies, and chassis which have successfully completed the electrical preconditioning shall be physically identified by a preconditioning stamp. The stamp shall be a "p" in a circle, and shall be marked on the preconditioned unit per the Group III requirements of MIL-M-13231. The stamp shall be affixed (1) to the chassis on the front panel near the name plate, (2) to the plug-in panel assemblies on the bottom side, and (3) to the Power Supply Assembly 4A1/5A1 on the bottom outside edge.

4.7.2 Bounce preconditioning (see 3.5.2). Each unit, in its non-operating condition, shall be placed in its normal operating position on the table of the package tester Type 1000 SC as made by the L.A.B. Corporation, Skaneateles, New York or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the bounces. The unit shall be subjected to this preconditioning for one minute prior to electrical preconditioning. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to electrical preconditioning.

4.8 Electrical tests.

4.8.1 Standard conditions. Unless otherwise specified, all tests shall be performed at 115 volts ±5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperatures, pressure and humidity per MIL-STD-810 unless otherwise specified.

4.8.2 Alignment for normal operation.

a. Apply power to equipment.

b. Align power supply DC voltages to hairline if meter indicates outside yellow region

c. Align TO RADIO XMTR and FROM RADIO XMTR signals to hairline on the TEST ALIGN meter.

d. Adjust CL control for maximum indication on the TEST ALIGN meter after placing the meter select switch to the SERV FAC placing the SERV SEL switch in the C position.

4.8.3 Standard test equipment. Standard test equipment shall be as indicated in test figures by manufacturer and model number. In all cases, equivalent substitutes may be used. Government Technical Representative approval is required prior to use of proposed substitute items. The equipment listed in Table VII may be used by the contractor as standard test equipment.

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TABLE VII STANDARD TEST EQUIPMENT

Designation	Function	Manufacture
Daven 12B	Transmission measuring set	Daven Div, Thomas A. Edison
GR1390B	Random noise generator	General Radio Co.
HP 200 CD	Wide-range Oscillator	Hewlett-Packard Co.
HP 330A	Distortion Analyzer	Hewlett-Packard Co.
HP 400L	Vacuum tube Voltmeter	Hewlett-Packard Co.
HP 5245L	Electronic counter	Hewlett-Packard Co.
HP 650A	Test Oscillator	Hewlett-Packard Co.
TEK R5403/D40	Dual Trace Oscilloscope	Tektronix Inc.
Clarostat 240C	Resistance Box	Clarostat Mfg. Co.
USVH	Frequency Selective Voltmeter	Rhode & Schwartz
Fluke 8120A	AC/DC Digital Voltmeter	Fluke, Inc.

4.8.4 Special test equipment. Special test equipment shall be indicated by name in test figures and followed by a description and block diagram of the equipment. Unless otherwise specified special test equipment shall be provided by the contractor subject to approval of the Government.

4.8.5 Ancillary Items. Ancillary items required for testing by this specification shall be furnished by the contractor in the quantities required to accomplish the testing herein.

4.8.5.1 Gaussian low pass filter. - The characteristics of the low pass gaussian response filter shall be as outlined in Table XVI.

TABLE XVI

Input Impedance with a termination of 51 ohms — 51 ohms ± 10%
Insertion Loss — 0.8 db maximum
Response:

Frequency	Attenuation
50kc (reference)	0 db
235 kHz	3.0 db ± 0.5 db
332 kHz	6.0 db ± 1.0 db
470 kHz	12.0 db ± 2.0 db
680 kHz	24.0 db ± 3.0 db
944 kHz	44.0 db ± 9.0 db

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4.8.5.2 Interconnecting cables. The interconnecting cables used with the Multiplexer TD-202( )/U are as follows:

4.8.5.3 Cable Assembly, Power Electrical, CX-1173( )/U. This assembly shall be fabricated per SC-DL-531015.

4.8.5.4 Cable Assembly, Radio Frequency CG-1040( )/U. This assembly shall be fabricated per SC-DL-531012.

4.8.5.5 Cable Assembly, Radio Frequency, CG-1049( )/. This assembly shall be fabricated per SC-DL-157329.

4.8.5.6 Cable Assembly, Special Purpose, CG-2438 ( )/TCC. This assembly shall be fabricated per SC-DL-531018.

4.8.5.7 Extender panel. The extender panel provides for operation of one panel with all components, controls, connections and test points accessible for troubleshooting. The extender panel shall be in accordance with applicable drawings.

4.8.5.8 Extender cable (4A1/5A1) Power Supply. The extender cable shall consist of a multiconductor cable with a connector at each end built in accordance with SM-D-530670. It shall be constructed for insertion between the chassis and the plug-in power supply subassembly and used for maintenance and alignment. This cable shall not be less than 18 inches in length or more than 20 inches in length.

4.8.6 Operational Test and Alignment

a. Connect equipment per figure 1.

b. Apply power to equipment.

c. Check all service facility readings.

d. Align power supply voltages to hairline of meter indicates outside the yellow region.

e. Align TO RADIO XMTR and FROM RADIO RCVR signals to give hairline indication on the test meter.

f. Adjust CL on 4A3/5A3 for maximum TEST ALIGN meter indication with the service facilities switch in the C position.

g. The requirements of 3.14 shall be met.

4.8.7 24 channel switchover test.

a. Connect equipment per Figure 2.

b. Align equipment for normal operation.

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c. Disconnect PCM IN 1 cable and compare PCM IN 2 signal versus TO RADIO XMTR signal.

d. Reconnect PCM IN 1 cable.

e. Disconnect PCM IN 2 cable and compare PCM IN 2 signal versus TO RADIO XMTR signal.

f. Reconnect PCM IN 2 cable.

g. Compare PCM IN 1 signal and PCM IN 2 signal versus TO RADIO XMTR signal.

h. The requirements of 3.4.4.1 shall be met.

4.8.8 Automatic phasing test.

a. Connect equipment per Figure 3.

b. Align equipment for normal operation.

c. Vary phasing of PCM IN 2 while monitoring audio channel.

d. Vary phasing of PCM IN 2 while monitoring PCM TO RADIO XMTR.

e. Requirements of 3.4.4.1 shall be met

4.8.9 Separation test.

a. Connect equipment per Figure 4.

b. Place Multiplexer TD-202( )/U under test mode switch in 24S position.

c. Align equipment for normal operation.

d. Place ADDRESS switch of send master Multiplexer TD-660B/G in SLAVE position. Note that both receive Multiplexers TD-660B/G go into frame alarm.

e. Return switch for normal operation.

f. Place ADDRESS switch of receive master Multiplexer TD-660B/G in in MASTER position. Note that this multiplexer goes into frame alarm.

g. Return switch for normal operation.

h. Place ADDRESS switch of receive slave Multiplexer TD-660B/G in MASTER position. Note that this multiplexer goes into frame alarm.

i. The requirements of 3.4.4.2 shall be met.

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4.8.10 Orderwire recovery test.

a. Connect the equipment per Figure 5.

b. Align equipment for normal operation.

c. Adjust PCM level FROM RADIO RCVR for 0.75v peak-to-peak.

d. Adjust oscillator level to provide for 200 mv peak to peak 1KHz sine wave at FROM RADIO RCVR.

e. Adjust orderwire level control for -4dbm out of Multiplexer TD-202( )/U.

f. Measure 1 KHz signal to distortion (S/D) ratio at orderwire terminals.

g. Measure signal to noise (S/N) ratio at orderwire terminals.

h. Remove PCM signal and measure 1 KHz signal to distortion (S/D) ratio at orderwire terminals.

i. Monitor orderwire output of Multiplexer TD-202( )/U and increase output level by clockwise adjustment of ORDERWIRE LEVEL control.

j. The requirements of 3.4.5.1 and 3.4.8.7 shall be met.

4.8.11 Traffic alarm test.

a. Connect the equipment per Figure 6.

b. Align equipment for normal operation.

c. Adjust PCM level FROM RADIO RCVR for 0.75v peak-to-peak.

d. Disconnect TO RADIO XMTR cable and note operation of TRAFFIC ALARMS and buzzer. Also note 1 KHz tone level both at the radio and the TD-202( )/U output.

e. Reconnect TO RADIO XMTR cable and turn ac power switch on Multiplexer TD-202( )/U "A" to off. Note 1KHz tone level at the TD-202( )/U output.

f. The requirement of 3.4.6.1 shall be met.

4.8.12 PCM signal test.

a. Connect equipment per Figure 7.

b. Align equipment for normal operation.

c. Measure PCM OUT 1 and PCM OUT 2 characteristics.

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d. Connect equipment per Figure 19.

e. Measure input impedance of PCM IN 1 and PCM in 2 at 100MHz

f. This requirement of 3.4.1 shall be met.

4.8.13 Video signal test.

a. Connect equipment per Figure 8.

b. Align equipment for normal operation.

c. Monitor TO RADIO XMTR signal on scope and note range of adjustments.

d. Vary input level and note correct operation over range of 0.5v to 3.0v peak-to-peak.

e. Connect equipment per Figure 20.

f. Measure input impedance of FROM RADIO RCVR at 10 Hz.

g. The requirements of 3.4.2 shall be met.

4.8.14 Timing signal test.

a. Connect equipment per Figure 9.

b. Align equipment for normal operation.

c. Measure TIMING OUT 1 and TIMING OUT 2 characteristics.

d. Connect equipment per Figure 21.

e. Measure input impedance of TIMING IN at 10 kHz.

f. The requirement of 3.4.3 shall be met.

4.8.15 Binary video detection test (low level)

a. Connect equipment per Figure 10.

b. Align equipment for normal operation.

c. Set PCM FROM RADIO RCVR level at 0.5v pp and adjust input level control for hairline indication on FROM RADIO RCVR position.

d. Set input peak signal to RMS noise ratio to 18 db.

e. Measure the BER.

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f. Requirements of 3.4.4.3 shall be met.

4.8.16 Binary video detection test (high level)

a. Connect equipment per Figure 10.

b. Align equipment for normal operation.

c. Set PCM FROM RADIO RCVR level at 0.5 pp and adjust input level control for hairline indication on FROM RADIO RCVR position.

d. Set input peak signal to RMS noise ratio to 23 db.

e. Measure the BER.

f. The requirements of 3.4.4.3 shall be met.

4.8.17 Biternary video detection test (low level)

a. Connect equipment per Figure 10.

b. Align equipment for normal operation.

c. Set PCM FROM RADIO RCVR level at 0.5v pp and adjust input level control for hairline indication on FROM RADIO RCVR position.

d. Adjust input peak signal to RMS noise ratio to 24 db.

e. Measure the BER.

f. Place mode selector switch of Multiplexer TD-202( )/U in 24S position.

g. Repeat c through e above.

h. Requirements 3.4.4.4 shall be met.

4.8.18 Biternary video detection test (high level)

a. Connect equipment per Figure 10.

b. Align equipment for normal operation.

c. Set PCM FROM RADIO RCVR level at 0.5v pp and adjust input level control for hairline inication on FROM RADIO RCVR position.

d. Set input peak signal to RMS noise ratio to 28 db.

e. The requirements of 3.4.4.4 shall be met.

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4.8.19 AC input test.

a. Connect the equipment for normal operation.

b. Align equipment for normal operation.

c. Perform test of channel S/N, 1KHz distortion, and output level stability on 3 audio channels, master and slave for each of the following voltages: 109v at 47.5 Hz; 121v at 47.5 Hz; 109v at 60 Hz; 115v at 60 Hz 121v at 60 Hz; 109v at 63Hz; and 121v at 63Hz.

d. The equipment shall meet specified performance at all times in accordance with 3.4.7.1.

4.8.20 Power supply output test.

a. Connect the equipment per Figure 13.

b. Align equipment for normal for normal operation.

c. Measure dc power supply voltage and ripple for +10, +4.5 -4.5 and -12 with the following ac input voltages: 109v at 47.5 Hz; 121v at 47.5 Hz; 109v at 60 Hz; 115v at 60 Hz; 121v at 60 Hz; 109v at 63 Hz and 121v at 63 Hz.

d. Determine that range of voltage adjustment meets requirements of Table I.

e. The requirements 3.4.7.2 and 3.4.7.3 shall be met.

4.8.21 Metering test.

a. Connect equipment per Figure 14.

b. Align equipment for normal operation.

c. Align +10, +4.5 and -4.5 supplies for hairline on TEST ALIGN meter.

d. Measure voltage of regulator outputs.

e. The requirements of 3.4.7.4 shall be met.

4.8.22 TO RADIO XMTR alignment test.

a. Connect equipment per Figure 15.

b. Align equipment for normal operation.

c. Place front panel selector switch on TO RADIO XMTR position and adjust OL control on panel 5A2 for hairline indication on the TEST ALIGN meter.

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d. Measure the peak to peak voltage of the TO RADIO XMTR signal.

e. The requirements of 3.4.8.4 shall be met.

4.8.23 Gaussian low pass filter test.

a. Connect equipment per Figure 16.

b. Align equipment for normal operation.

c. Measure insertion loss of the filter at the frequencies specified in Table II.

d. Measure filter input impedance.

e. The requirements of 3.4.8.2 shall be met.

4.8.24 Service condition test (Temperature, Humidity).

a. Connect equipment per Figure 17

b. Align equipment for 24 channel operation. (This step is performed only once during entire temperature or humidity test cycle).

c. Measure The BER for both directions.

d. Measure the orderwire signal to distortion (S/D) ratio.

e. Measure the order wire signal level.

f. Talk-listen on the orderwire.

g. The requirements of 3.4.5.1, 3.4.5.2 and 3.4.4.4 shall be met.

4.8.25 Service condition test (Dripproof, Vibration, Bounce, Shock, Low Pressure and Dust).

a. Connect equipment per Figure 18.

b. Align equipment for 24 channel operation.

c. Measure power supply DC voltages of unit under test.

*d. Signal, talk and listen on system orderwire in both directions.

e. The requirements of 3.4.5.2 and 3.5.6.2 shall be met.

* DO NOT SIGNAL AT THE TRANSPORTATION ALTITUDE IN THE ELEVATION TEST.

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4.9 Electromagnetic compatibility. (see 3.11) The equipment shall be tested for complaince to the requirements of MIL-STD-461, Notice 4, Table A-1, for Class CE equipment. The test methods and procedures described in MIL-STD-462, Notice 3 shall be employed. All electromagnetic compatibility tests shall be performed with the equipment simulating normal operation. The frequency range of the following tests are modified as follows: RE02 -14 khz to 1000 Mhz and RS03 -14 Khz to 1000 Mhz.

a. Connect equipment per Figure 18.

b. Align equipment for 12 channel operation.

c. Perform radiated and conducted emission tests on the equipment in accordance with MIL-STD-462 Notice 3 and the approved EMC/EMI test plan.

d. Perform radiated and conducted susceptibility tests on the equipment in accordance with MIL-STD-462 Notice 3 and the approved EMC/EMI test plan while monitoring the following:

(1) Listen on one audio channel.

(2) Listen on orderwire.

e. The requirements of 3.10 shall be met.

4.10 Inspection for dimensional interchangeability. The gage points identified on SC-GL-323241 shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11.

4.11 Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252 for conformance with MIL-STD-454 and the requirements of 3.3 and 3.15. An inspection shall be performed to verify complaince with those portions of 3.13 (Safety Engineering) which can be determined visually.

4.12 Electrical subassembly test. (See 3.12) Multiplexer TD-202( )/U and plug-in subassemblies, including running spare panel assemblies, shall be tested in accordance with their respective test requirement specification drawings listed in 3.12.

4.13 Quality conformance inspection of preparation for delivery. Preparation for delivery hsall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

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4.14 Cover Assembly, Case, Electronic Equipment. The cover shall be constructed in accordance with, and meet the requirements of, SM-D-423893. This cover shall be used in the Dripproof Test (see 3.6.6), Salt Fog Test (see 3.6.7), and Dust Test (see 3.6.8).

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. JR-1-1-ØØ-OO-NS-X-ED-Ø-OO-A.

5.1.2 Level B. JR-1-1-ØØ-OO-NS-X-ED-Ø-OO-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. Multiplexer TD-202( )/U operates in ATACS Shelter assemblages including AN/TRC-108, -109,-110,-117,-151 and -152.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, Level B or commercial preservation and packing (see section 5).

c. When the rough handling test is required.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature required, test tools and running spares.

g. First Article Inspection (see 3.1).

h. Length of cables (if any are required).

6.3 Definitions.

63.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (Hypha).

6.3.2 Exfoliation. Exfoliation is corrosion along the grain boundries of the metal resulting in the peeling or separating, or both, of successive

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layers of the metal. The appearance resembles loose pages or onion skin peeling.

6.3.3 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.3.4 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.3.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.3.6 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.4 Group C inspection. (see paragraph 4.5.3.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. (see paragraph 4.5.3) Initial shipments of end items will be contingent only upon successful completion of all Group C tests.

6.5 Location of operational inspection. (see paragraph 4.8.6) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TD-202/U. The contractor should apply for nomenclautre in accordance with the applicable clause in the contract.

6.7 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.8 Environmental. Environmental pollution prevention measures are contained

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in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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Preparing Activity
ARMY-CR
Project No. 5805-A302

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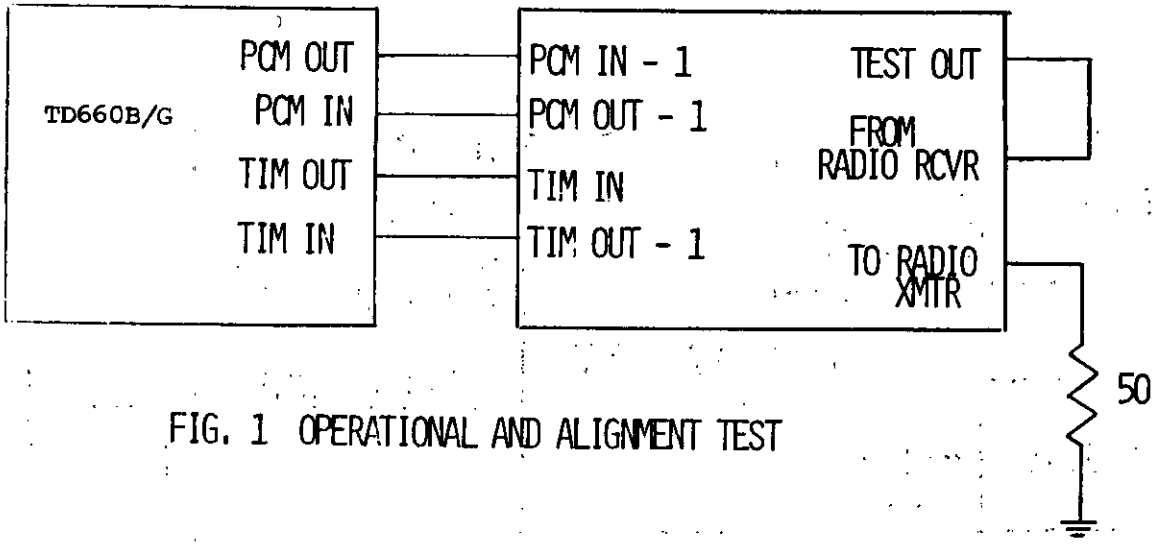


FIG. 1 OPERATIONAL AND ALIGNMENT TEST

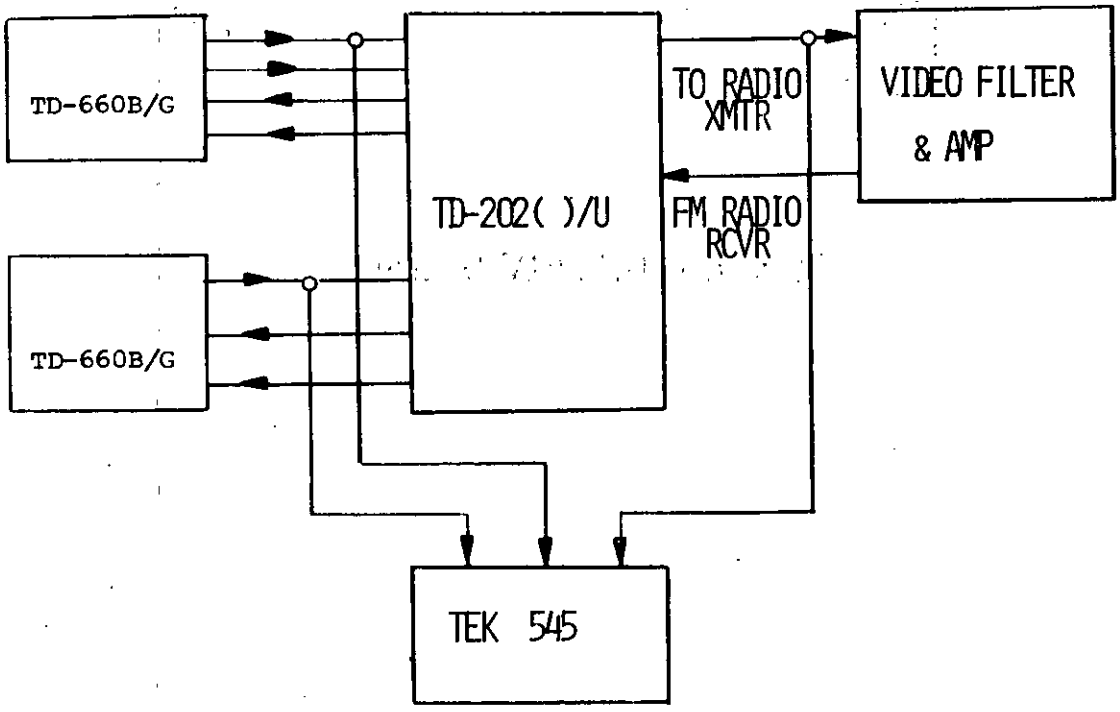


FIG. 2 CHANNEL SWITCHOVER TEST

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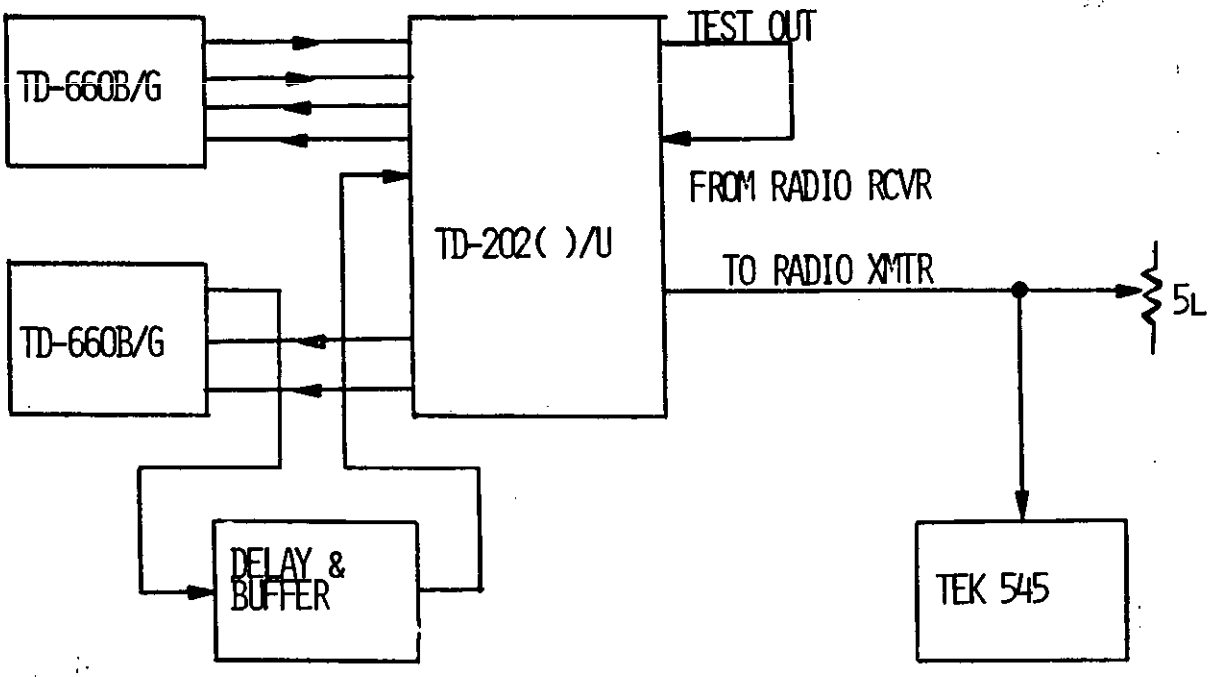


FIG 3 AUTOMATIC PHASING TEST

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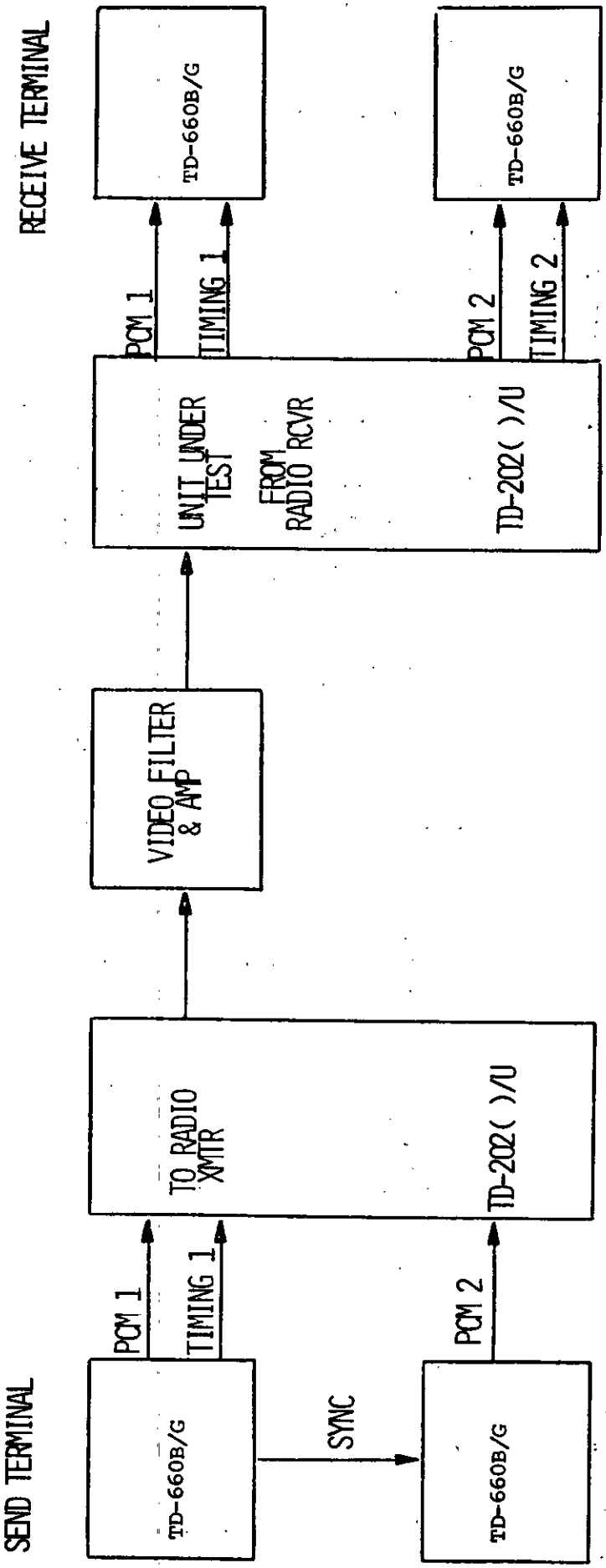


Fig 4 SEPARATION TEST

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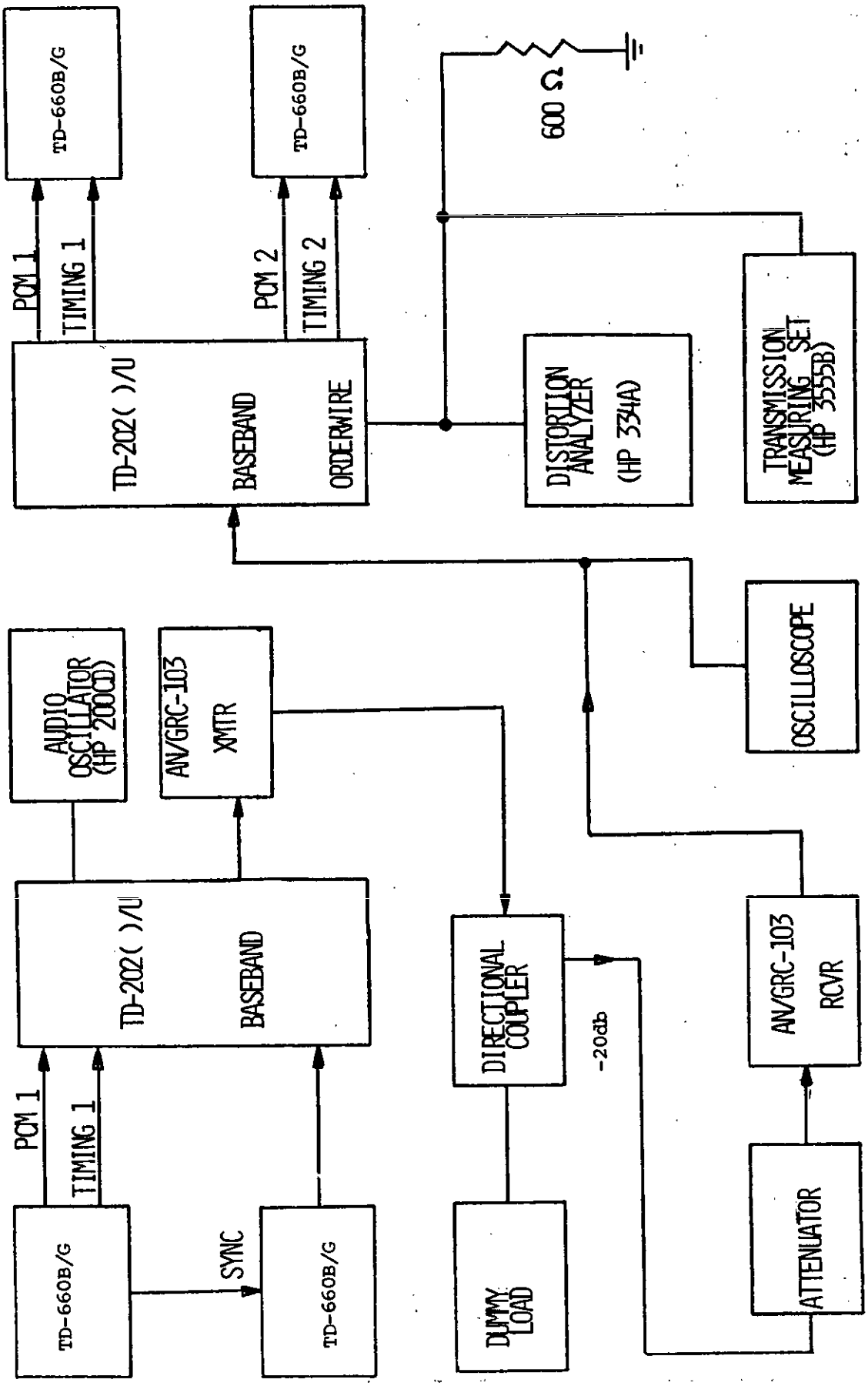


Figure 5 ORDERWIRE RECOVERY TEST

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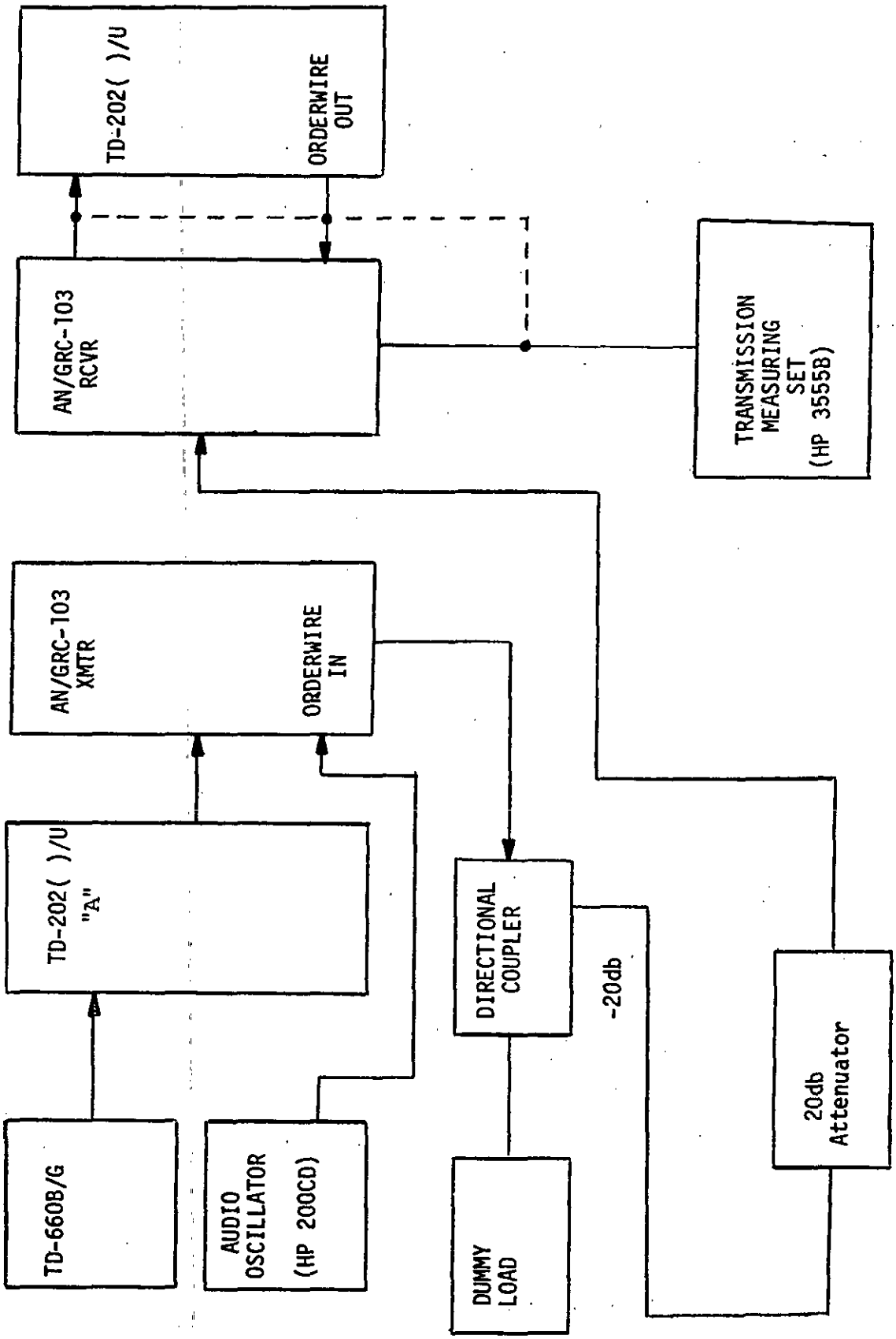


Fig 6 TRAFFIC ALARM TEST

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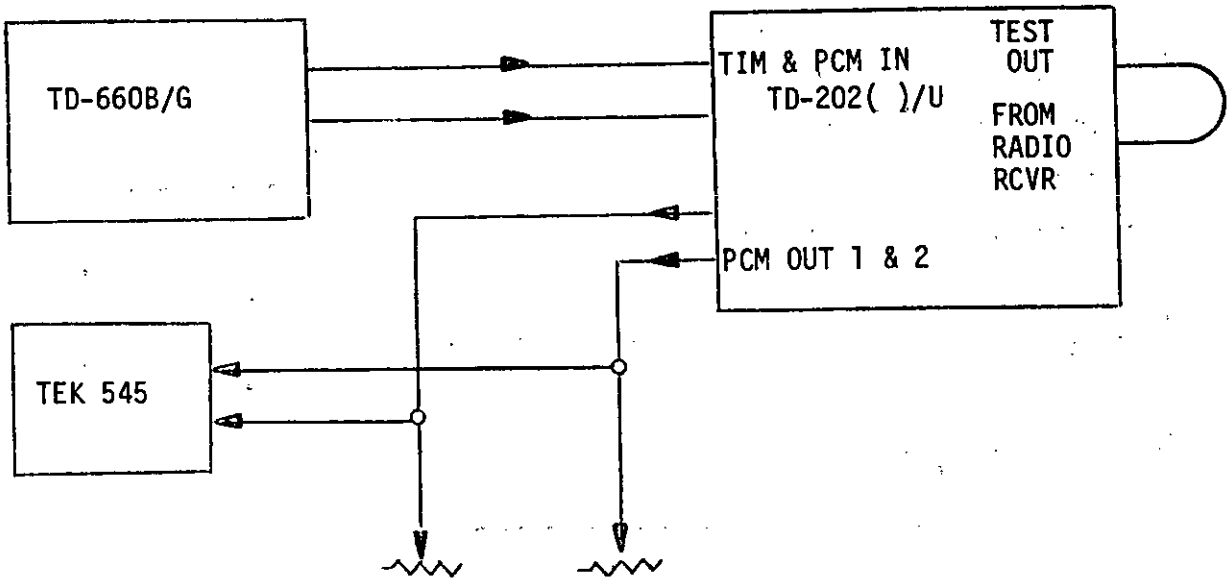


Fig 7 PCM SIGNAL TEST

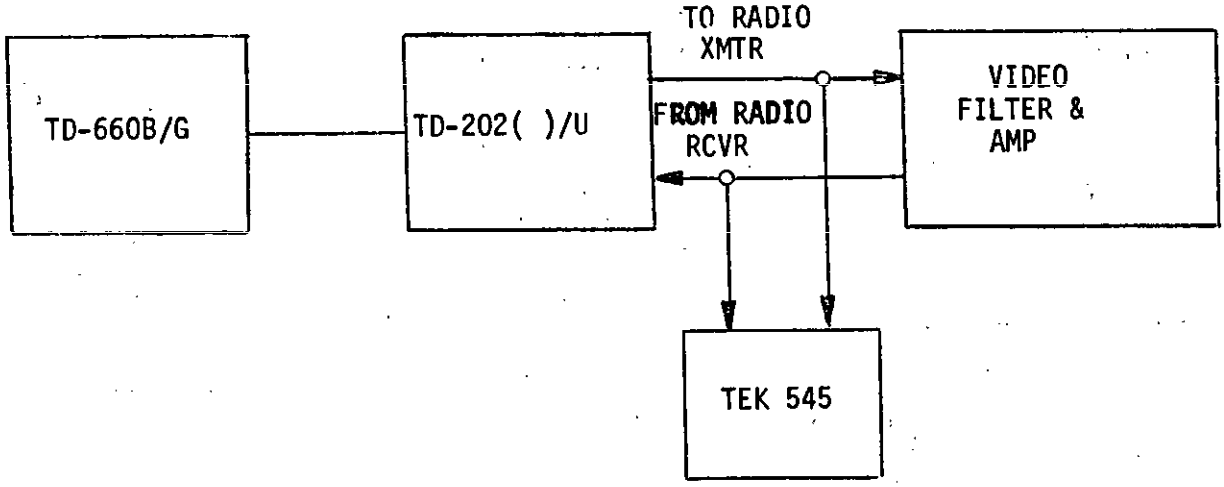


Fig 8 VIDEO SIGNAL TEST

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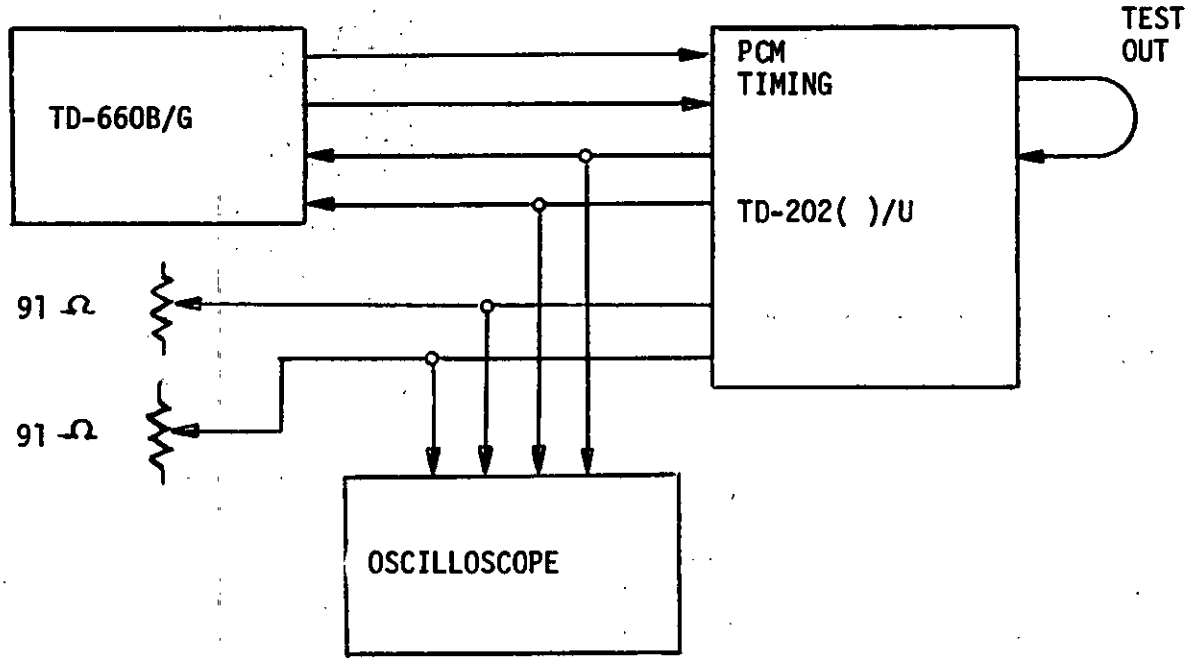
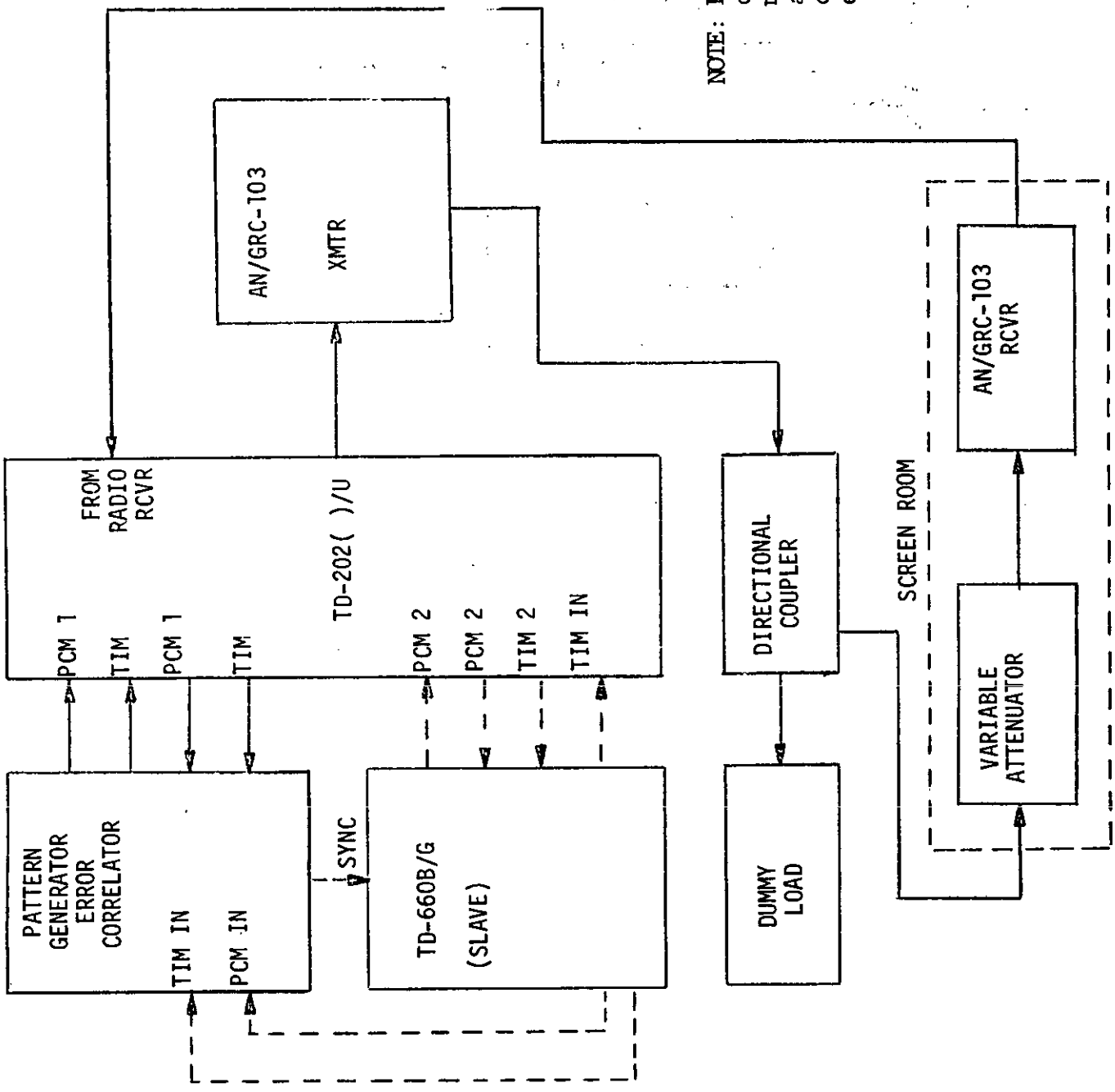


Fig 9 TIMING SIGNAL TEST

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NOTE: Dashed Lines are to be disregarded for 12-channel (binary) operation, and observed for 24-channel (biternary) operation.



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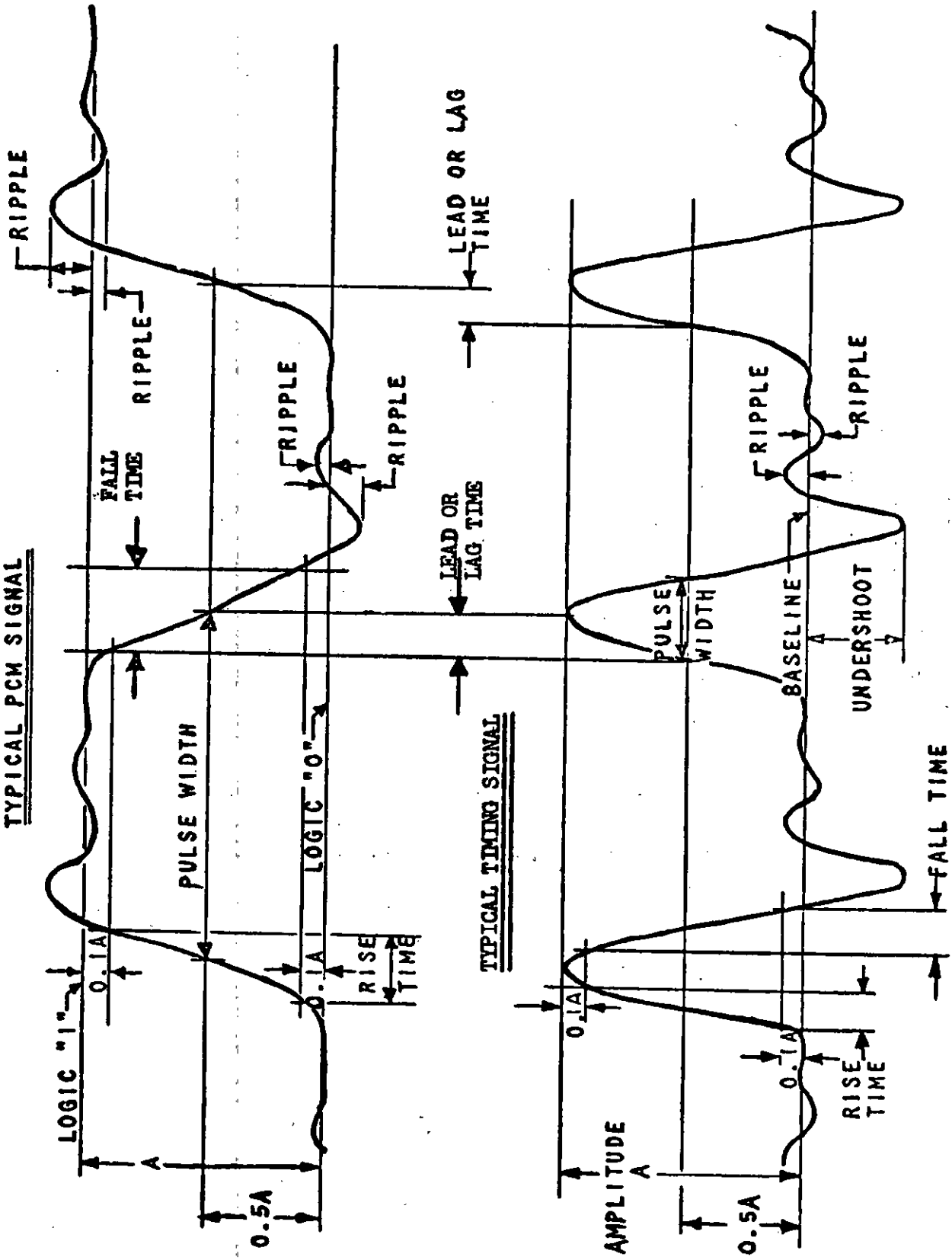


FIG 11 SIGNAL PARAMETERS

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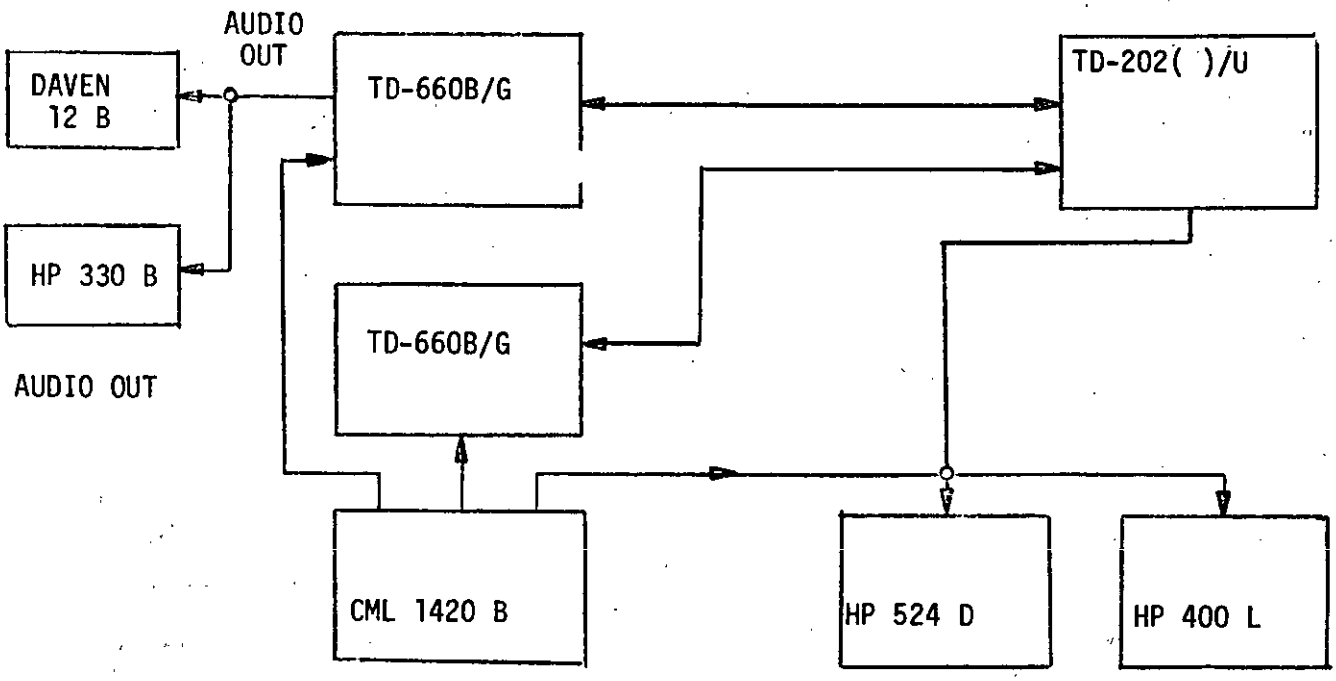


Fig 12 AC INPUT TEST

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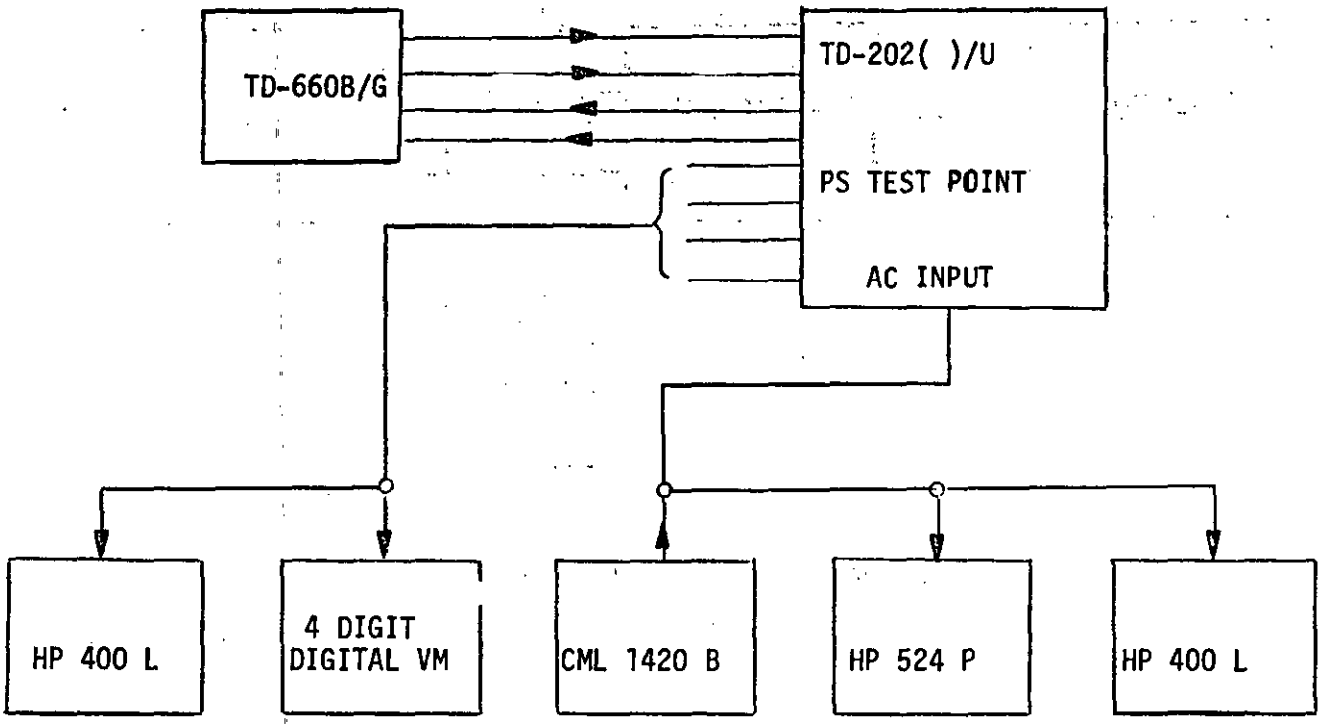


Fig 13 POWER SUPPLY OUTPUT TEST

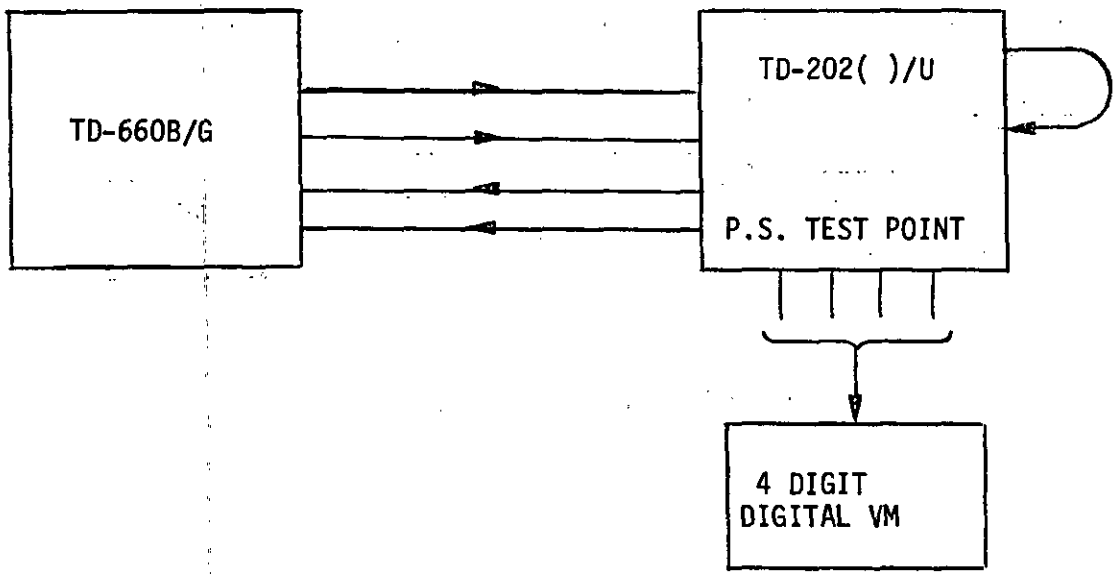


FIG 14 METERING TEST

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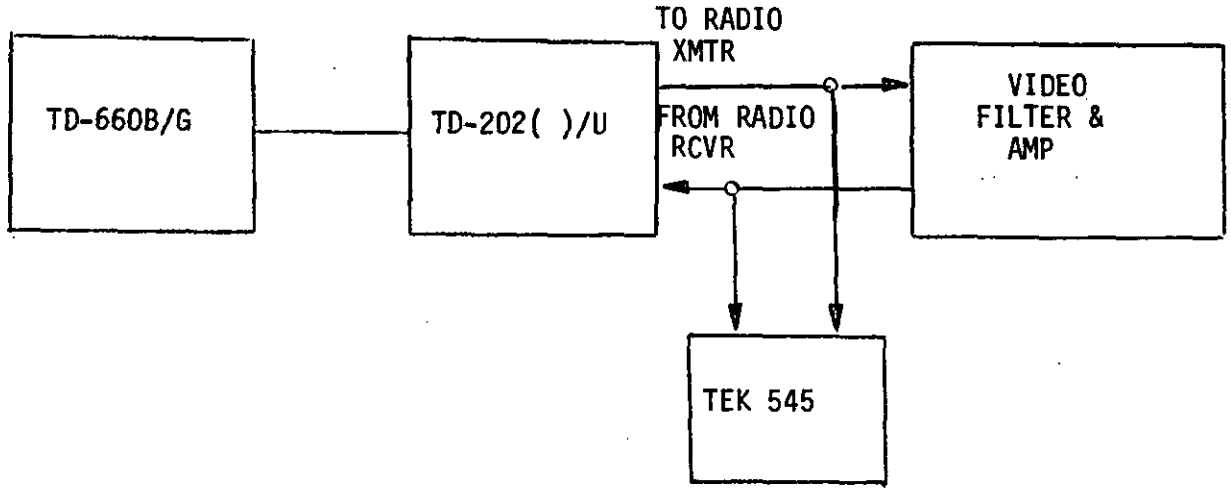


Fig 15 TO RADIO XMTR ALIGNMENT TEST

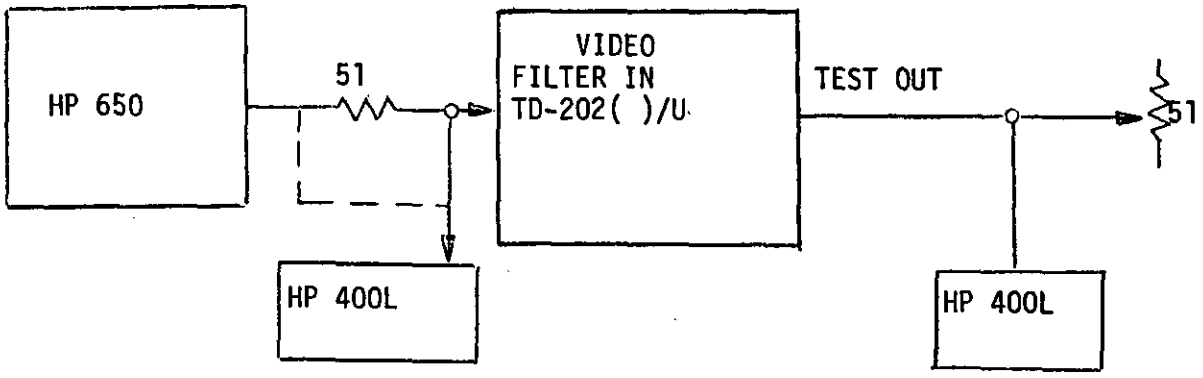


Fig 16 GAUSSIAN LP FILTER TEST

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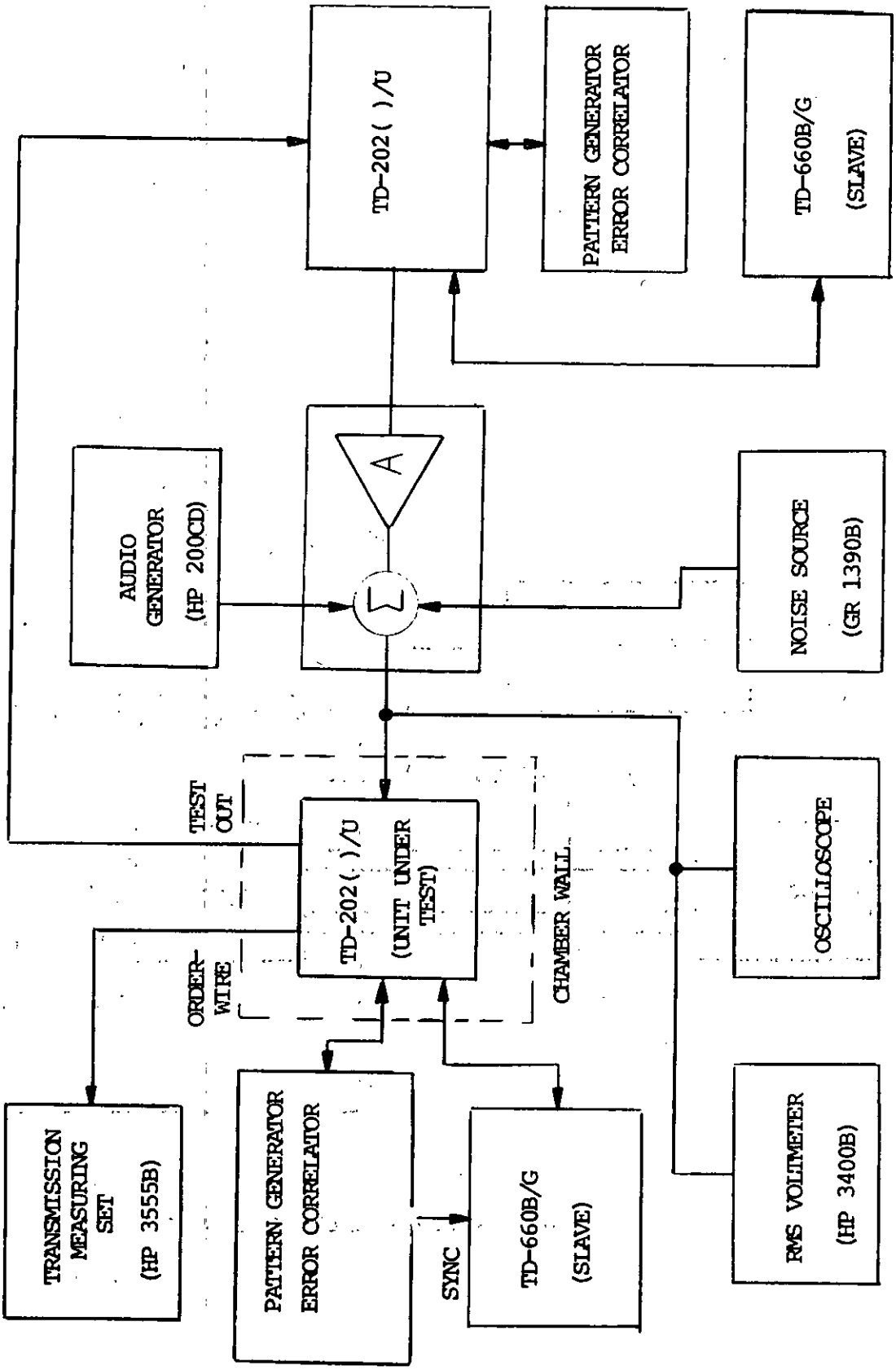


FIG 17 SERVICE CONDITION TEST

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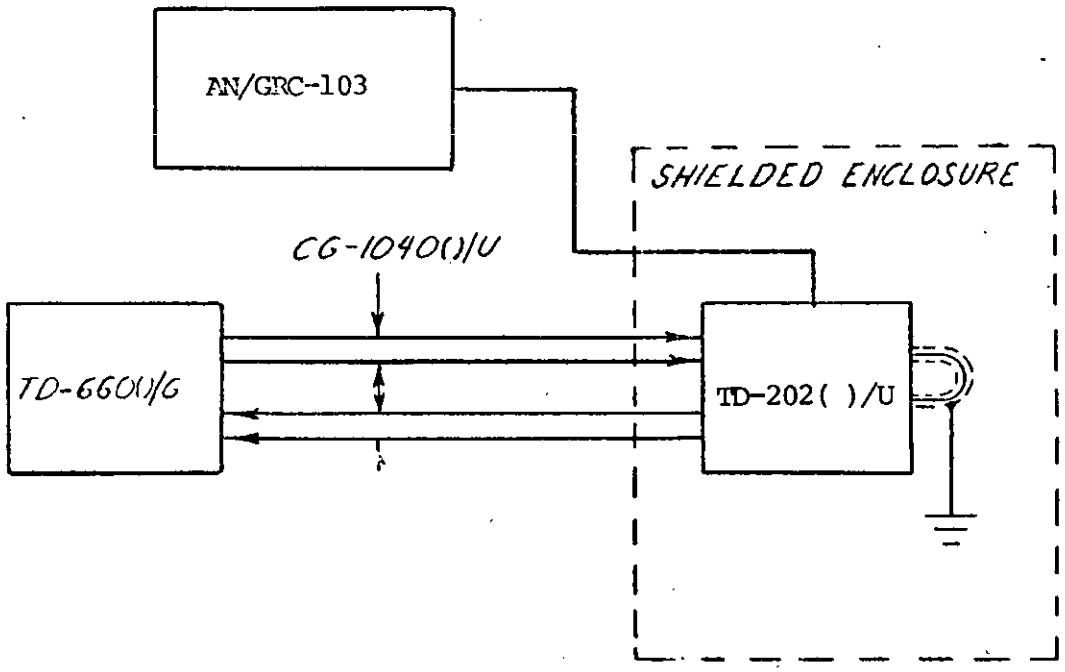


FIG 18 ELECTROMAGNETIC COMPATABILITY TEST

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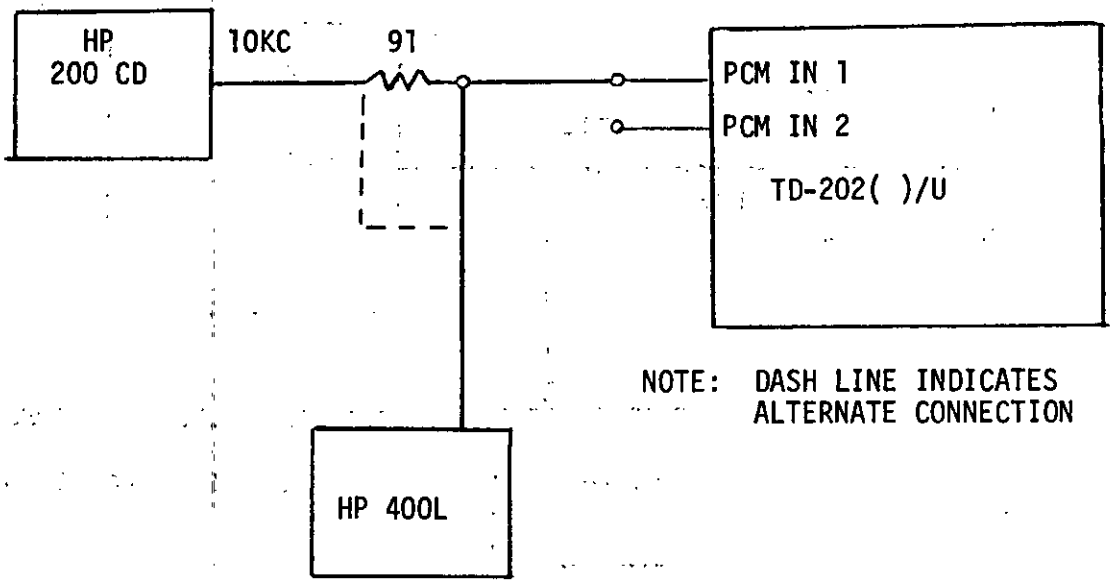


Fig 19 PCM SIGNAL TEST

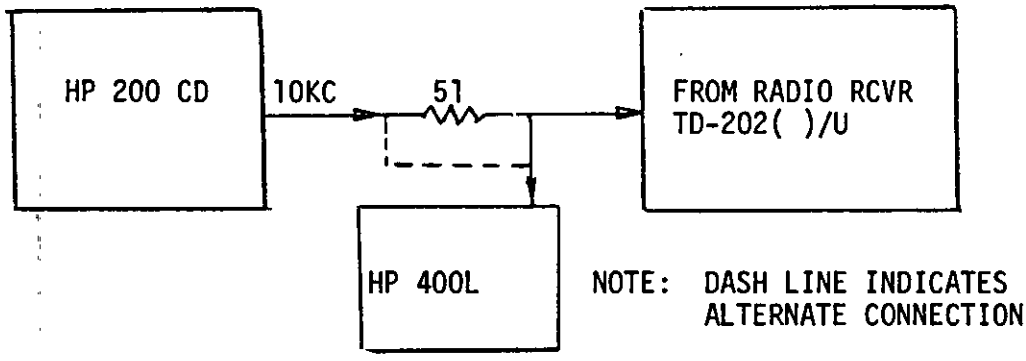


Fig 20 VIDEO SIGNAL TEST

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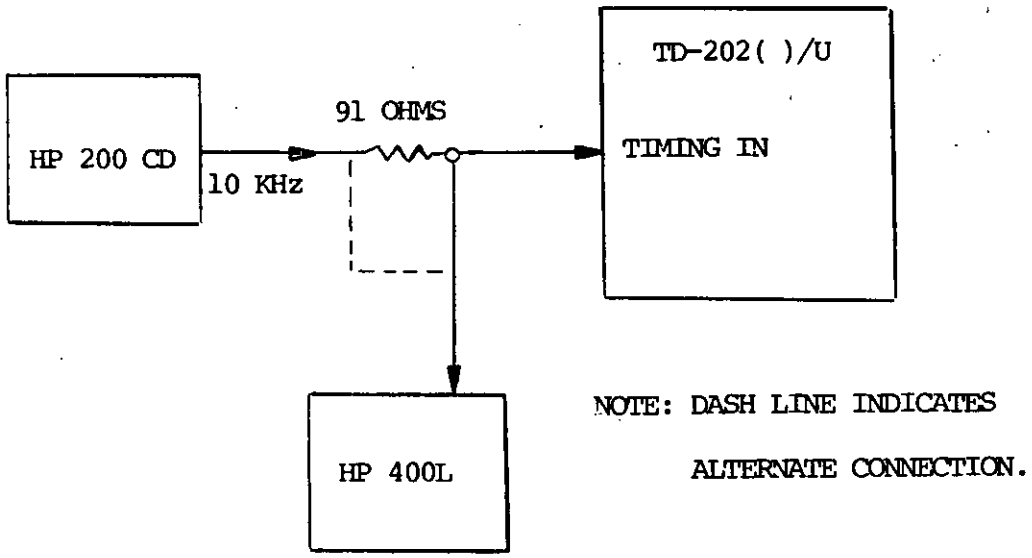


FIG 21 TIMING SIGNAL TEST

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

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